Robust Deepfake Detection Using Lightweight Deep Learning Models

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ABSTRACT Reliable detection of synthetic facial images requires assessing both classification performance and sensitivity to image degradation. This study compares MobileNetV3-Small and ResNet18 for real-versus-synthetic facial-image classification using the balanced RVF10K dataset. The principal contributions are a stage-separated architecture comparison, a matched-input corruption study with class-specific error analysis, which is a measure of accuracy and efficiency and a fixed-target analysis that checks explanation consistency. The experimental framework uses transfer learning splits data into training and validation sets and evaluates the outcomes on 3,000 held-out images, five corruption categories, computational measurements, and gradient-based visual explanations. At the comparable earlier fine-tuned stage, MobileNetV3-Small achieves 83.13% clean accuracy and ResNet18 achieves 83.87%. MobileNetV3-Small uses 1.52 million parameters, compared with 11.18 million for ResNet18, and has approximately 4.46 times the throughput in the recorded central-processing-unit evaluation loop. Under strong image compression and resolution reduction, the compact model retains higher threshold-based detection performance, whereas ResNet18 achieves higher accuracy under the tested Gaussian-noise conditions. Both models exhibit severe losses in fake-image recall under strong blur. Extended MobileNetV3-Small fine-tuning raises held-out accuracy to 85.63% and the area under the receiver operating characteristic curve to 0.9355; matching extended ResNet18 test results are unavailable. Pairwise explainability analysis provides diagnostic evidence of spatial consistency when applying compression to the images. A useful trade-off between accuracy and efficiency was found with notable limitations depending on the conditions. Conclusions are limited to the recorded dataset, checkpoints and evaluation conditions.

INDEX TERMS Deepfake detection, explainable artificial intelligence, image classification, lightweight neural networks, robustness, synthetic faces, transfer learning.

I. INTRODUCTION

DEEPFAKE technology uses generative deep learning techniques to create or edit facial images so it can be difficult to authenticate digital media. As deepfake technology is easily accessible, there are concerns around misuse of identity, misinformation, privacy and trust in digital evidence. Therefore, developing an automated deepfake detection system becomes an important information security issue. However, detecting deepfake images using a clean dataset does not guarantee the detector will behave reliably when images are compressed, noisy, blurry, altered in brightness or reduced in resolution. Therefore, when evaluating a deepfake detector, it is important to ensure it behaves reliably by considering the transformed conditions [1]–[3].

In terms of computational efficiency, larger neural networks can offer powerful representations but will come with increased memory and processing requirements. MobileNetV3-Small provides an efficient architecture [4] to investigate small facial image detectors. ResNet18 is used as a residual network baseline [5]. When comparing the architectures, it is important to consider the error rates, number of parameters and measured execution conditions, as well as aggregate accuracy.

In this study, both networks have been evaluated using the RVF10K balanced real-versus-synthetic facial image dataset [6], [7]. This will be limited to static image classification and will not aim to develop a universal deepfake detector for videos with swapped faces, temporal tampering or audiovisual deepfakes. The initial training, previous fine-tuning and extended fine-tuning results have been separated due to different validation and checkpoint strategies being used in the experiments. This separation prevents measurements from different model states from being presented as a single final-model evaluation.

A. RESEARCH QUESTIONS

1) **RQ1:** How is MobileNetV3-Small and ResNet18 compare in terms of clean accuracy and class-specific errors at the comparable earlier fine-tuned stage?
2) **RQ2:** How is JPEG compression, Gaussian blur, Gaussian noise, brightness variation, and resolution reduction affect the two detectors?
3) **RQ3:** What efficient accuracy trade-off is observed in parameter count and recorded CPU evaluation throughput?
4) **RQ4:** What improvement is supported by extended fine-tuning, and what limits the updated comparison between architectures?
5) **RQ5:** What do Grad-CAM visualizations and paired heatmap similarities reveal about explanation consistency under JPEG compression?

Each question is answered in one corresponding paragraph in Section V.

B. RESEARCH CONTRIBUTIONS

The contributions of this study are as follows.

- A comparison of MobileNetV3-Small and ResNet18 reported separately for the initial, earlier fine-tuned, and extended checkpoints.
- A corruption study of 38 model-condition evaluations over five corruption families, with class-specific error analysis.
- A joint report of parameter count, CPU evaluation throughput, and clean accuracy.
- A fixed-target Grad-CAM consistency analysis of 20 paired images per model, with all image-level records reported.

This study provides a holistic evaluation with transparent results. This does not provide any novel architectures, corruption operators or explainability algorithms. There are no claims made around generalisation across datasets or unseen generators, or deployment reliability.

C. ORGANIZATION

Section II reviews relevant detection, robustness, and explanation research. Section III describes the workflow, architectures, training stages, and evaluation protocol. Section IV presents the experimental analysis. Section V answers the research questions, superseded by limitations and conclusions.

II. RELATED WORK

A. DEEPFAKE DETECTION

Wang et al. [1] provide a reliable perspective on deepfake detection. They highlight transferability, interpretability and robustness as important issues within the field. Pei et al. [2] provide a survey on deepfake generation and detection, including datasets, metrics, and representative approaches. These studies motivate evaluation beyond a single clean-data accuracy. Tanfoni et al. [8] investigate transfer-learning approaches for facial deepfake classification. Pretrained feature extractors provide a practical starting point for this task, but their adaptation must be evaluated under the target data and processing conditions.

B. EFFICIENT ARCHITECTURES

MobileNetV3 was designed for efficient visual inference through hardware-aware architecture design and lightweight operations [4]. ResNet introduced residual connections that support optimization of convolutional networks [5]. These families provide the basis for comparing the compact MobileNetV3-Small with ResNet18. The present study evaluates shared data preparation and metric definitions, while explicitly reporting differences between training stages.

C. ROBUSTNESS AND EXPLAINABILITY

Common-corruption benchmarking [3] allows analysis of how image degradation impacts classification. Bharati et al. [9] discuss explainable detection for forensics. Grad-CAM produces class-discriminative visualisations based on the gradients of convolutional feature maps [10]. The robustness metrics and heatmap consistency are separate observations. A visualisation acts as diagnostic evidence and not proof the prediction is correct or relies upon the highlighted pixels.

D. DETECTION BENCHMARKS

The wider literature covers fully synthesized faces, identity replacement, facial attribute changes, and expression manipulation. Mirsky and Lee [11] provide a review on creating and detecting deepfake images. Tolosana et al. [12] provide categorisation for facial image manipulation. Verdoliva [13] discusses deepfake problems within visual media forensics and limitations to verification tools. This is an important consideration for this experiment as a classifier trained on authentic and fully generated images will not necessarily act as a detector for different manipulated facial imagery.

FaceForensics++ [14] provides a benchmark for multiple facial manipulation methods. It considers compression so is relevant to the motivation behind testing the corrupt conditions. Celeb-DF [15] provides more challenging synthetic celebrity videos. DFDC [16] provides a large face-swap video dataset. These resources show why benchmark scope and manipulation type matter when designing an experiment. They aren't usef for testing or training in notebook but are useful for external evaluation. The published detection scores have not been used for the RVF10K dataset.

E. GENERALIZATION

Wang et al. [17] investigate how well a deepfake detector will generalise across CNN-based image generators. Preprocessing and augmentation were considered. Ojha et al. [18] study the more difficult transition across generative-model families and use pretrained representations for broader fake-image detection. Together, these studies motivate separating within-dataset performance from unseen-generator generalization. The present clean and corruption experiments address the former setting and controlled transformations of its images;

they do not reproduce the cross-generator protocols of these papers.

Face X-ray [19] uses evidence of image blending to investigate generalization across face-forgery techniques. Its assumptions are particularly relevant to composited facial manipulations, and should not be transferred without testing to entirely generated faces. F3-Net [20] instead investigates frequency-aware clues, using complementary image decomposition and local frequency statistics. This offers a possible direction for investigating the sensitivity to blur and compression observed here, but it does not prove that the present models rely on the same frequency cues or would obtain comparable improvements.

Multi-attentional deepfake detection [21] combines spatial attention with fine-grained texture information, while self-blended images [22] create synthetic training examples intended to encourage more general forgery representations. These approaches show alternatives to adapting a conventional global binary classifier. They are relevant future baselines, not implemented components of the current MobileNet and ResNet experiments. A fair comparison would require the same data, split, preprocessing, and evaluation policy rather than comparing published headline accuracies from different datasets.

DeepfakeBench [23] specifically addresses inconsistent data processing, experimental settings, and evaluation protocols in detector comparisons. Its emphasis on standardized benchmarking supports the stage-specific reporting adopted in this paper. The earlier and extended runs differ in validation preprocessing, checkpoint selection, and training budget. These differences are disclosed so that the table of results is not mistaken for an architecture-only ablation. Adding further baselines would be most informative after a common implementation and evaluation protocol is fixed.

F. DATASETS AND ARCHITECTURES

The StyleGAN work of Karras et al. [24] introduces a style-based generator and the FFHQ facial-image dataset. It provides background for the real-image source described by the RVF10K documentation [6], [7]. This provenance citation does not establish that every fake image in the local dataset was produced by a particular StyleGAN checkpoint. Generator-specific claims would require explicit metadata for the supplied files, beyond the real/fake directory labels inspected in the notebook.

MobileNetV2 [25] develops inverted residuals and linear bottlenecks for efficient visual models, providing architectural context for the later MobileNetV3 family [4]. EfficientNet [26] studies balanced scaling of network depth, width, and input resolution as another route to accuracy-efficiency trade-offs. These network families are useful to understand why the number of parameters and runtime have been measured. MobileNetV2 and EfficientNet have not been evaluated within this notebook. Their inclusion in the literature does not expand the set of experimentally tested architectures.

G. OPTIMIZATION AND VALIDATION

The Adam optimiser [27] has been used for the training code. This gives an indication of the algorithm being used. The learning rate, weight decay, scheduler and checkpoint selection have all been specific to this implementation. No optimiser ablation has been performed so we cannot say that Adam is better than other optimisation methods.

Guo et al. [28] look at probability calibration within neural networks. Post-processing methods have been considered. This is important to know if the detector will perform well with useful score ordering but poor performance at a particular decision threshold. However, calibration error, reliability diagrams and temperature scaling have not been measured within this notebook. The results have shown differences in AUC vs recall which will be useful for future calibration and decision point analysis but do not prove the detector is miscalibrated. They also do not validate a threshold correction.

Adebayo et al. [29] show why visual appeal is not enough when evaluating saliency techniques. Their study introduces model- and data-randomisation checks. This gives justification for the Grad-CAM overlay and cosine similarity results to be seen as diagnostic observations. These randomisations have not been implemented so the results have been shown based on the exact visualisations and paired similarities from JPEG. The explanation analysis distinguishes spatial consistency from prediction correctness.

The experiments have been performed using PyTorch [30]. The imperative programming style has been used for training, evaluating and visualising using gradients. This does not indicate that the original paper has verified the package versions or CPU timings. This literature shows that small detectors can be investigated with controlled corruption and explainability. However, generalisation across datasets and reliability during deployment requires further experimentation.

III. METHODOLOGY

Figure 1 shows the methodology used to compare MobileNetV3-Small and ResNet18 for real-versus-synthetic image classification. The workflow covers data splitting, preprocessing, transfer learning, checkpoint selection, clean evaluation, corruption testing, and explanation analysis. Both networks have the same image splits, classes, input size and metrics. However, training-stage differences have been documented. The previous and extended experiments do not use the same validation strategy.

A. DATASET

The RVF10K dataset contains 10,000 balanced facial images. 7,000 images have been provided in the training directory and 3,000 images have been provided in the validation directory [6], [7]. The training directory contains 3,500 images per class and the validation directory contains 1,500 images per class. The real-image source is associated with the FFHQ collection whilst the fake class contains synthetic faces. This dataset has been used for real-versus-synthetic image

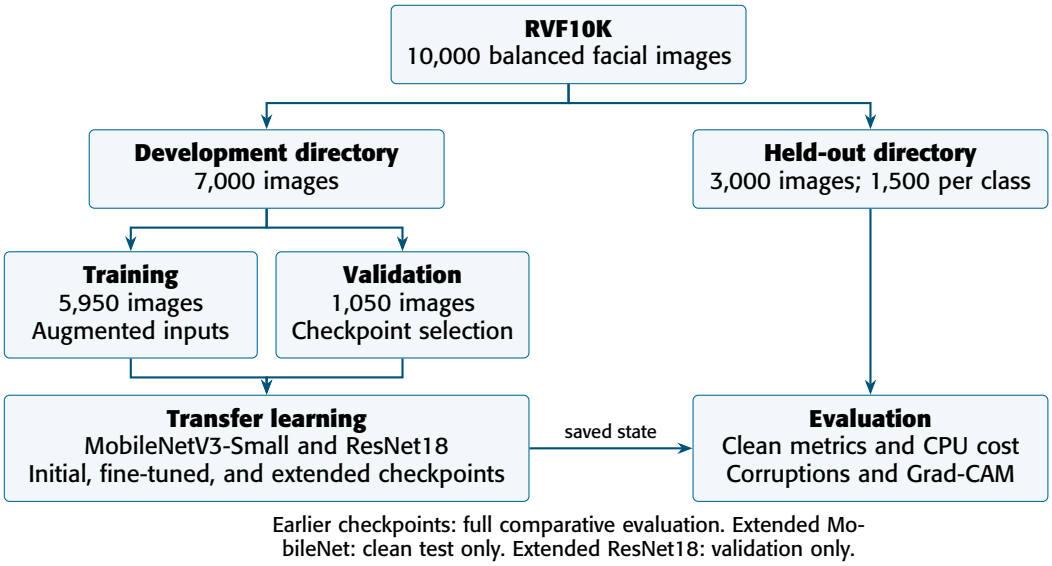


FIGURE 1. Experimental workflow. The supplied 3,000-image directory is used for held-out evaluation. Corruption and explanation results belong to the earlier fine-tuned checkpoints; the extended ResNet18 record contains internal validation only.

classification rather than as a universal benchmark for every deepfake technique. To make claims around specific generators more information is required from the metadata.

Real images are labelled as 0 and fake images as 1. The fake images are the positive class when training and evaluating. This determines the interpretation of precision, recall, false positives, false negatives and receiver operating characteristic (ROC) analysis. The classes have been clearly defined to avoid default directory loading ordering.

B. DATA PARTITIONING

The data has been split using stratified random sampling with a seed of 42. 15% of the 7,000 images have been split to create an internal validation set. This has resulted in 5,950 images for training and 1,050 images for validation. There are 2,975 images per class for training and 525 images per class for validation. Both architectures have used the same split indices. The `valid` directory containing 3,000 images has been provided as the held-out test directory. This directory does not provide gradient updates.

The extended experiments have used separate dataset instances for training and validation, both referencing the same source directory. The training data has been randomly augmented using stochastic augmentation. The validation data has been deterministically preprocessed. The previous cells have linked the validation subset to the augmented training data. The epoch choice has been based on the internal validation. The held-out directory has been inspected at multiple points during development. This makes this an exploratory study. The saved record does not establish identity-disjoint partitions or duplicate screening.

C. PREPROCESSING

The inputs have been converted to RGB and resized to 224×224 pixels. The floating point tensors have been normalised using the ImageNet channel statistics:

$$x'_c = \frac{x_c - \mu_c}{\sigma_c}, \tag{1}$$

where $\boldsymbol{\mu} = [0.485, 0.456, 0.406]$ and $\boldsymbol{\sigma} = [0.229, 0.224, 0.225]$. The training data has been randomly augmented with horizontal flips, rotation up to 10° and mild colour jitter affecting contrast, brightness, and saturation. This introduces variability in appearance but maintains the image class. The extended validation and clean test have been resized, converted to tensors and normalised. No random augmentation has been applied. During robustness evaluation, the source image has been corrupted before being preprocessed by the model. This avoids defining distortions inadvertently in normalized tensor space.

D. MODEL ARCHITECTURE

Figure 2 explains the two pretrained backbones and their binary outputs. MobileNetV3-Small uses lightweight inverted residual operations and channel attention within its feature extractor [4]. Its original final classification output is replaced by a single linear output unit. The instantiated binary model contains 1,518,881 parameters. In extended fine-tuning, the last three feature blocks and classifier are trainable, corresponding to 1,236,513 parameters; earlier blocks remain frozen.

ResNet18 supplies a conventional residual-network comparison [5]. Its feature extractor uses residual blocks, followed by global pooling and a fully connected classifier. The final output is again replaced by an unit. The final output is

the thing left and the resulting model contains 11,177,025 parameters. Further fine-tuning trains the final stage, `layer4`, and the classifier, which totals 8,394,241 trainable parameters. These diagrams describe the instantiated architecture families; they do not imply that every backbone layer is updated at every training stage. Figure 3 gives the layer-level structure of both networks and shades the regions that are unfrozen during fine-tuning. The per-block parameter counts shown there add up to the stated trainable totals: $294{,}096 + 294{,}096 + 56{,}448 + 591{,}873 = 1{,}236{,}513$ for MobileNetV3-Small and $8{,}393{,}728 + 513 = 8{,}394{,}241$ for ResNet18.

E. CLASSIFICATION OBJECTIVE

For an image x, the network produces logit z. The fake-class probability and default decision are

$$p(y = 1 \mid x) = \sigma(z) = \frac{1}{1 + \exp(-z)}, \qquad \hat{y} = \mathbf{1}[\sigma(z) \geq 0.5]. \tag{2}$$

Training uses binary cross-entropy with logics. For labels $y_i \in \{0, 1\}$ and a batch of N samples,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log \sigma(z_i) + (1 - y_i) \log \big(1 - \sigma(z_i)\big) \Big]. \tag{3}$$

The implementation uses both cross-entropy and sigmoid in a numerically stable operation. Labels are floating-point values compatible with the one-logit output. Probabilities support both threshold-based metrics and score-based ROC analysis; loss itself is not substituted for a final performance metric.

F. OPTIMIZATION

The initial training helper freezes feature-extractor parameters and fits the binary classifier using Adam [27], learning rate 10^{-3}, and weight decay 10^{-4}. The earlier procedure uses validation accuracy for scheduling and checkpoint selection, with early-stopping patience of three. Calling `model.train()` can still update batch-normalization running statistics in frozen modules, so parameter freezing alone does not guarantee an unchanged feature-extractor state. The earlier fine-tuning runs and their saved learning curves are reported separately in Section IV.

Extended fine-tuning restores each earlier fine-tuned model state and initializes a new Adam optimizer with learning rate 10^{-5} and weight decay 10^{-4}. It uses a validation-loss plateau scheduler, a ten-epoch cap, and early-stopping patience of three. Because optimizer state is initialized afresh, this is continued fine-tuning rather than an exact resumption of the prior optimization trajectory. The previous checkpoint is preserved. Frozen feature modules are explicitly kept in evaluation mode so their batch-normalization statistics do not drift.

G. CHECKPOINT SELECTION

For the extended experiments, the model state is saved whenever internal validation loss improves, and the best saved state is restored after training. The final epoch is not selected automatically. Epoch-level training loss, accuracy, validation loss, validation accuracy, and timing describe the learning trajectory. Decreasing training loss alone does not establish improved generalization, and development accuracy is not an independent held-out result. The earlier experiments select by validation accuracy and use augmented validation inputs, whereas the extended experiments minimize deterministic validation loss. These explain why results are grouped by checkpoint stage.

H. EVALUATION METRICS

Clean evaluation uses deterministic preprocessing on all 3,000 held-out images without gradients or weight updates. True positives (TP) are fake images classified as fake; false positives (FP) are real images classified as fake; true negatives (TN) are real images classified as real; and false negatives (FN) are fake images classified as real. The threshold-based metrics are

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \tag{4}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{5}$$

$$F_1 = \frac{2\,\text{Precision}\,\text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

Undefined metric cases are handled consistently, including reported zeros when fake detections are absent. ROC-AUC evaluates ranking across thresholds using predicted probabilities. Confusion matrices expose the balance between false alarms and missed fake images. A useful AUC need not imply satisfactory recall at threshold 0.5. Any alternative threshold requires selection on development data followed by separate evaluation.

I. CORRUPTION EVALUATION

Five corruption families are evaluated without retraining: JPEG compression, Gaussian blur, additive Gaussian noise, brightness variation, and resolution reduction followed by upscaling. JPEG qualities 90, 70, 50, and 30 use all 3,000 images. Blur radii 1, 2, 3, and 5; noise standard deviations 5, 15, 25, and 40; brightness factors 0.6, 0.8, 1.2, and 1.4; and resolution scales 0.75, 0.50, and 0.25 use a fixed stratified subset of 500 images, with 250 per class. This gives 19 conditions per model and 38 model-condition evaluations.

Each corruption is applied independently to the original image, and both models receive the same source samples and settings. Gaussian noise is defined on the 0–255 pixel scale and clipped to that range. The later evaluator derives a deterministic seed from the image index and severity, so both models receive the same noise realization. Resolution reduction uses bilinear downsampling and upsampling to the

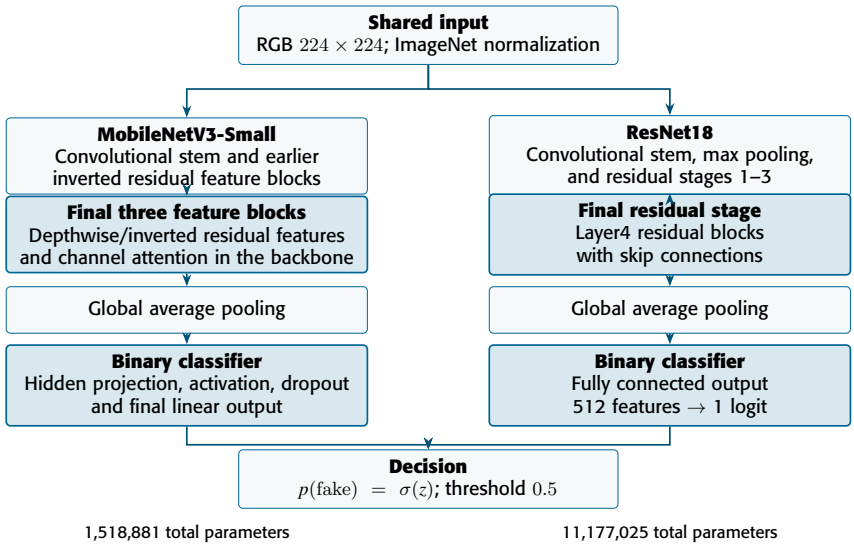


FIGURE 2. Schematic binary-classification architectures. Both models use ImageNet initialization and a single output logit. Highlighted adaptation regions indicate the later feature blocks trained in the extended experiments. The displayed parameter counts are measured for the binary models.

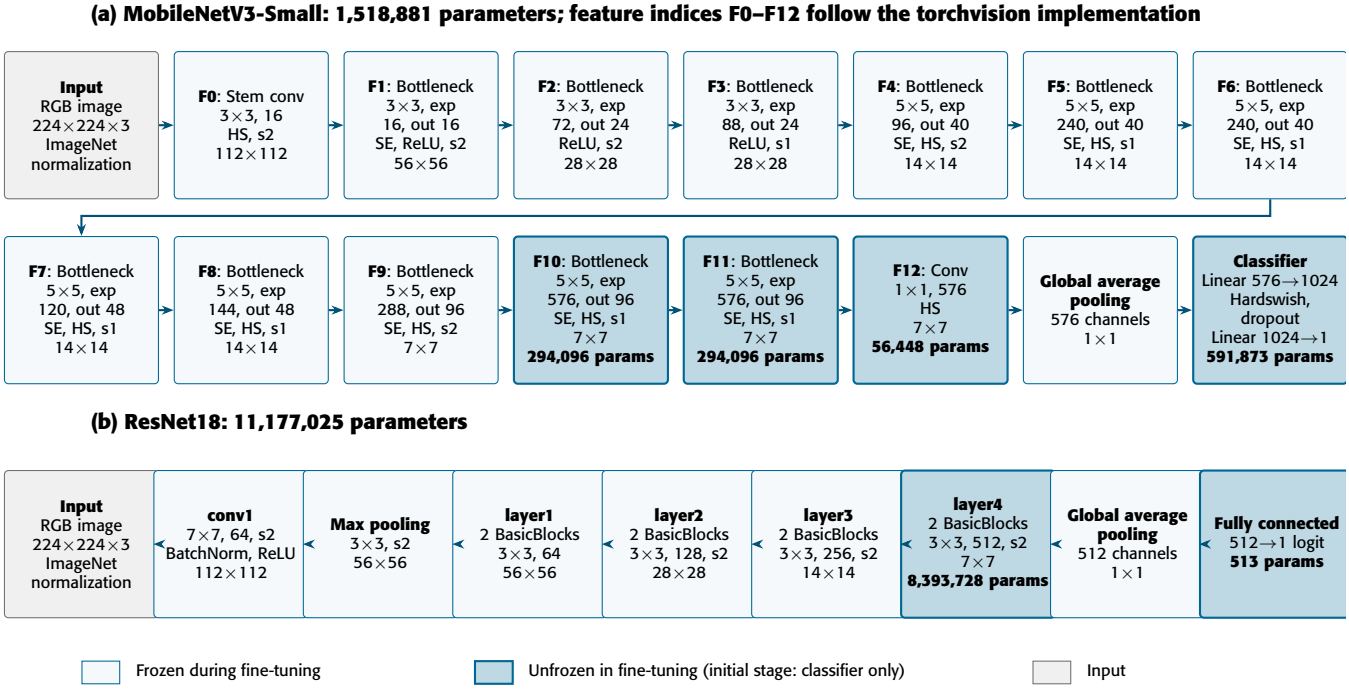


FIGURE 3. Layer-level structure of the two detectors for a 224×224 input. Bottleneck entries give kernel size, expansion width (exp), output channels (out), squeeze-and-excitation (SE), activation (ReLU or Hardswish, HS), and stride (s). Resolutions are output feature-map sizes. Shaded blocks are unfrozen during fine-tuning and show their parameter counts.

original dimensions before normal preprocessing. A resolution scale therefore refers to the source dimensions rather than one assumed fixed intermediate pixel size.

For a metric M, a matched clean-to-corrupt change is defined as

$$\Delta M = M_{\text{corrupt}} - M_{\text{clean}}. \tag{7}$$

The clean and corrupted terms must use identical images. Full-set JPEG measurements satisfy this comparison. The saved record does not contain a clean baseline for the exact 500-image subset, so the other categories report absolute corrupted metrics and changes between evaluated settings. They do not use the full-set clean score as a substitute subset baseline. All reported corruption results belong to the earlier fine-tuned checkpoints, not the extended models.

J. GRAD-CAM

Grad-CAM [10] visualizes regions associated with a selected class score using the final MobileNet feature block and the last block of ResNet18's `layer4`. For feature map A^k and target score s^c, the channel importance weight is

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial s^c}{\partial A_{ij}^k}, \tag{8}$$

where Z is the number of spatial locations. The coarse localization map is

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{9}$$

The map is resized and overlaid on the source image. The binary target uses logit z for fake and $-z$ for real. In the qualitative examples, the target is each model's predicted class, so an incorrect real prediction produces a real-target explanation. The available four-image comparison contains fake examples only; one row uses different targets across architectures. These overlays are diagnostic and do not establish correctness, explanation faithfulness, or causal localization.

K. EXPLANATION CONSISTENCY

Twenty image indices are sampled without replacement using seed 42. Both earlier fine-tuned models use the same clean/JPEG-quality-50 image pairs. Within each model-image pair, the clean predicted class is the target for both Grad-CAM maps, preventing a target change from being conflated with a spatial explanation change. For flattened maps u and v, similarity is

$$\text{CosSim}(u, v) = \frac{u^\top v}{\|u\|_2 \, \|v\|_2}. \tag{10}$$

The implementation returns zero for a zero-norm map. Mean, sample standard deviation, minimum, maximum, and all image-level values are reported. A high similarity indicates similar spatial patterns in this representation; it does not establish faithful explanations or unchanged predictions. The saved table contains clean probabilities, clean predicted classes, and similarities, but not compressed probabilities, compressed classes or ground-truth labels. Nothing can be said about how often the predictions change or if consistent maps correlate with correct decisions.

L. COMPUTATIONAL MEASUREMENTS

The total number of parameters has been counted from the instantiated binary networks. The time taken to evaluate on the CPU has been measured. This includes loading the data, preprocessing, inference and collecting results. Throughput has been calculated by dividing the number of images by the time taken. Amortised cost has been calculated by dividing the time taken by the number of images. This provides information on how the run performed. These are evaluation-loop measurements rather than forward-pass-only latency.

TABLE 1. Data partitions and experimental sample counts. Per-class counts for the internal split follow from the balanced stratified split.

Partition / analysis	Images	Real / fake
Training	5,950	2,975 / 2,975
Internal validation	1,050	525 / 525
Clean test and JPEG	3,000	1,500 / 1,500
Other corruptions	500	250 / 250
JPEG explanation pairs	20 per model	Not stratified

No information has been provided on the processor or memory specifications, an explicit warm-up, variability between runs or energy usage.

JPEG has been run on individual images. The 500-image corruption evaluator has been run in batches of 32. The time taken to run the corruption loop has included transforming the images. This cannot be directly compared between families as it is not an indication of how fast the network is. The extended training has been run using 4 CPU threads. No matching extended inference benchmarks have been saved. The number of parameters, time taken and prediction performance have been interpreted together. These comparisons remain tied to their stated checkpoints and measurement conditions.

M. EXPERIMENTAL CONTROL

The evaluation records split sizes, class mapping, preprocessing, initialisation, trainable-layer configurations, optimiser settings, selection criteria, corruption severities, and explanation targets. The original experiment has been restarted and has out of order counters. The dataset, trained checkpoints and standalone predictions have not been provided with the paper. The manuscript only provides evidence of what has been saved and does not claim to reproduce training from scratch. The experiment has only been run on one dataset, the held-out directory has been inspected multiple times, sampling has been limited, and some extended measurements are missing. This has been considered in Section VI.

IV. ANALYSIS

A. EXPERIMENTAL SETUP

The experiment has been saved with training logs, printed metrics, data tables and visualisations. The results have been split into initial training, previous fine-tuning and extended training. The numerical results and original figures have been saved. No extra training or inference has been performed.

The dataset has been confirmed to contain 7,000 images in the development source and 3,000 images in the held-out directory. Both directories have the same number of real and fake classes. The data has been split using stratified sampling with 15% of the data used for internal validation. This has given 5,950 images for training and 1,050 images for validation. All binary metrics have been based on real $= 0$, fake $= 1$. This means the precision, recall and F1 values are relating to fake image detection unless a class-specific result has been identified.

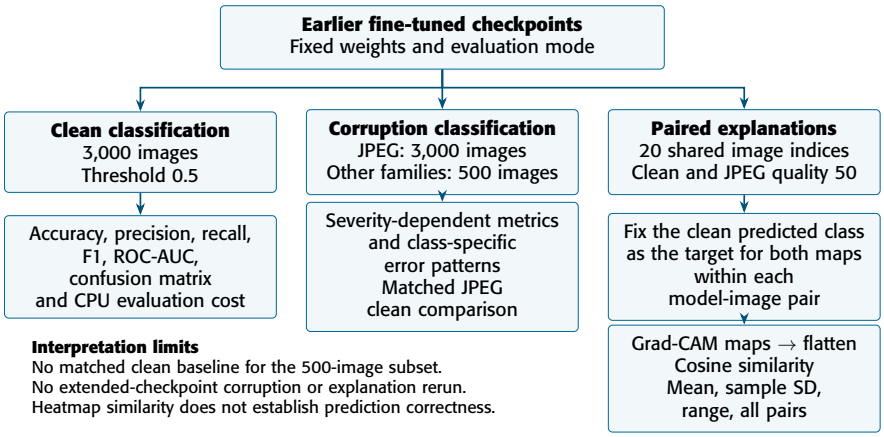


FIGURE 4. Evaluation branches and pairwise explainability. The corruption metrics and heatmap similarities provide different information. The clean predicted class has been fixed for each clean/compressed explainability pair. No compressed predictions have been saved in the similarity table.

TABLE 2. Checkpoint-stage evidence. M denotes MobileNetV3-Small; R denotes ResNet18.

Stage	M evidence	R evidence
First classifier	10 epochs; clean test	7 epochs; clean test
Previous fine-tuning	4 epochs; clean test	2 epochs; clean test
Strength / CAM	Earlier FT checkpoint	Earlier FT checkpoint
Further fine-tuning	10 epochs; clean test	10 epochs; validation only

TABLE 3. Preliminary MobileNet run; training-only outputs, not held-out results.

Epoch	Training loss	Training accuracy
1	0.5505	0.7247
2	0.4890	0.7620
3	0.4612	0.7820
4	0.4411	0.7936
5	0.4010	0.8118

TABLE 4. Initial MobileNetV3-Small training. L = loss; A = accuracy. Epoch 9 has the selected validation accuracy.

Ep.	Train L	Train A	Val L	Val A	min
1	0.5523	0.7084	0.4643	0.7867	1.00
2	0.4971	0.7625	0.4517	0.7810	0.86
3	0.4691	0.7761	0.5670	0.7143	0.86
4	0.4376	0.7929	0.4106	0.8105	0.86
5	0.4192	0.8029	0.4114	0.8086	0.86
6	0.3947	0.8212	0.4053	0.8076	0.92
7	0.3845	0.8230	0.3798	0.8210	0.93
8	0.3675	0.8378	0.4062	0.8133	0.96
9	0.3702	0.8323	0.3717	0.8352	0.97
10	0.3704	0.8371	0.3907	0.8267	1.01

The complete earlier fine-tuned comparison provides clean metrics, confusion matrices, ROC curves, CPU evaluation times, 38 model-condition corruption results, and 40 explanation-consistency records. The extended MobileNetV3-Small run supplies ten training epochs and a clean held-out evaluation. The extended ResNet18 run supplies ten training epochs but no corresponding saved held-out evaluation. Consequently, its validation accuracy is never substituted for test accuracy.

The implementation uses PyTorch [30]. The software outputs report CPU execution and CUDA unavailable. The displayed package versions are PyTorch 2.14.1+cu130, Torchvision 0.29.1+cu130, and pandas 3.0.6. These are reported notebook values rather than independently checked installation metadata. The extended cells set four CPU threads. The processor model, memory capacity, and a repeated timing protocol are not included in the record.

B. TRAINING

A preliminary five-epoch MobileNet experiment precedes the main comparison. Training loss falls from 0.5505, to 0.4010 while training accuracy rises from 0.7247 to 0.8118. A fresh pretrained MobileNetV3-Small is subsequently instantiated for the actual experiment, so these five preliminary epochs are not added to the main model's training history.

The main initial training freezes the feature-extractor parameters and fits the binary classification head. The common helper uses Adam [27] with learning rate 0.001 and weight decay 0.0001, an accuracy validating scheduler, and patience of three for early stopping. Calling `model.train()` still allows batch-normalization running statistics to change in frozen modules; parameter freezing alone does not freeze those statistics. This differs from the explicit frozen-module evaluation mode used during the extended runs.

MobileNet's selected validation accuracy is 0.8352 at epoch 9, compared with 0.7867 after the first epoch. Validation temporarily deteriorates at epoch 3, where loss reaches 0.5670 and accuracy falls to 0.7143. Recovery in subsequent epochs supports examining the full trajectory rather than treating the final epoch as automatically optimal. Epoch 10 has lower validation accuracy than epoch 9, so the helper restores the earlier selected state.

ResNet18 reaches its best initial validation accuracy of 0.7533 at epoch 4. Three subsequent epochs do not exceed

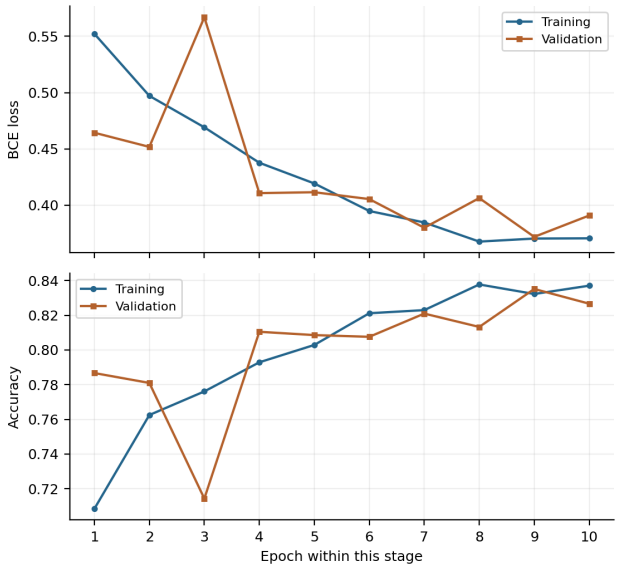


FIGURE 5. Initial MobileNet training and validation trajectories reconstructed from the saved epoch logs. Validation inherits stochastic augmentation in this stage.

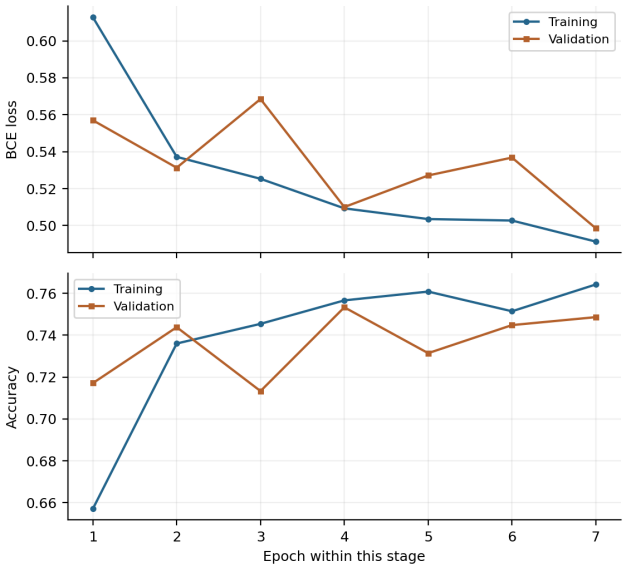


FIGURE 6. Initial ResNet18 trajectories reconstructed from saved logs. The selected epoch is determined by validation accuracy, not the lowest validation loss.

TABLE 5. Initial ResNet18 training. Epoch 4 is selected; early stopping occurs after epoch 7.

Ep.	Train L	Train A	Val L	Val A	min
1	0.6128	0.6571	0.5570	0.7171	2.94
2	0.5372	0.7360	0.5313	0.7438	3.06
3	0.5253	0.7454	0.5685	0.7133	3.01
4	0.5093	0.7566	0.5100	0.7533	3.05
5	0.5035	0.7608	0.5271	0.7314	2.96
6	0.5027	0.7514	0.5368	0.7448	2.99
7	0.4913	0.7642	0.4985	0.7486	2.96

this value, triggering early stopping after epoch 7. Epoch 7 has a lower validation loss of 0.4985 than epoch 4's 0.5100, but the helper monitors accuracy, explaining why epoch 4 remains selected. This illustrates that the selected checkpoint can change when the selection criterion changes.

Earlier fine-tuning enables the final three MobileNet feature blocks and its classifier, and ResNet18's final residual stage and classifier. The requested learning rate is reduced to 0.00001. MobileNet is allowed five epochs with patience two but stops after four; ResNet18 is run for two epochs with patience one. These unequal fine-tuning budgets are part of the recorded comparison and limit attributing every difference solely to architecture.

MobileNet reaches validation accuracy 0.8371 at fine-tuning epoch 2. Later epochs achieve lower training loss but do not exceed this selection score. ResNet improves from validation accuracy 0.8124 to 0.8171 over its two fine-tuning epochs. Because this earlier validation pipeline applies random augmentation, individual epoch values are subject to augmentation variability in addition to model changes.

The notebook execution counters are not fully sequential: for example, the displayed MobileNet unfreezing cell has

TABLE 6. Earlier MobileNet fine-tuning. Epoch 2 is selected by validation accuracy.

Ep.	Train L	Train A	Val L	Val A	min
1	0.3535	0.8469	0.3717	0.8305	1.10
2	0.3437	0.8526	0.3706	0.8371	1.04
3	0.3254	0.8674	0.3552	0.8362	1.02
4	0.3205	0.8649	0.3624	0.8295	1.06

TABLE 7. Earlier ResNet18 fine-tuning. Epoch 2 is selected.

Ep.	Train L	Train A	Val L	Val A	min
1	0.4203	0.8089	0.4190	0.8124	3.58
2	0.3680	0.8375	0.3957	0.8171	3.68

TABLE 8. Initial classifier-stage clean evaluation on 3,000 images. AUC uses probabilities; the other metrics use threshold 0.5.

Model	Acc.	Prec.	Recall	F1	AUC
Mobile	0.8217	0.8631	0.7647	0.8109	0.9117
ResNet	0.7687	0.7669	0.7720	0.7694	0.8404

a later execution number than its fine-tuning call. Several cells also contain pasted console output inside code cells. The numerical tables here reproduce saved results; a clean, top-to-bottom rerun with the original data and checkpoints would be needed to establish complete execution provenance. Repeated displays of the same saved tables are consolidated rather than counted as new experiments.

C. INITIAL EVALUATION

The initial MobileNet model records 82.17% accuracy and an ROC-AUC of 0.9117, compared with 76.87% and 0.8404

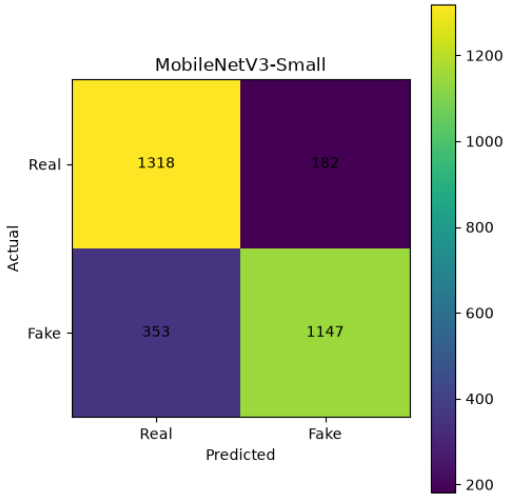


FIGURE 7. Confusion matrix for the initial MobileNet checkpoint. Rows are actual class and columns are predicted class.

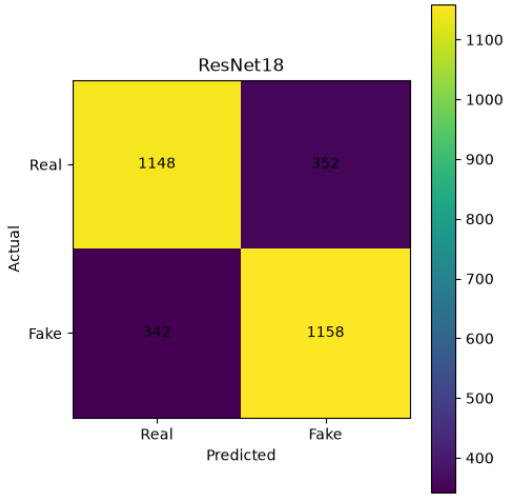


FIGURE 8. Confusion matrix for the initial ResNet18 checkpoint.

TABLE 9. Initial class-specific reports retained from the notebook. Support is 1,500 per class.

Model / class	Precision	Recall	F1
Mobile / Real	0.7887	0.8787	0.8313
Mobile / Fake	0.8631	0.7647	0.8109
ResNet / Real	0.7705	0.7653	0.7679
ResNet / Fake	0.7669	0.7720	0.7694

TABLE 10. Earlier fine-tuned clean evaluation on 3,000 images. These checkpoints are also used for the saved robustness and explanation studies.

Model	Acc.	Prec.	Recall	F1	AUC
Mobile	0.8313	0.8045	0.8753	0.8384	0.9173
ResNet	0.8387	0.8307	0.8507	0.8406	0.9198

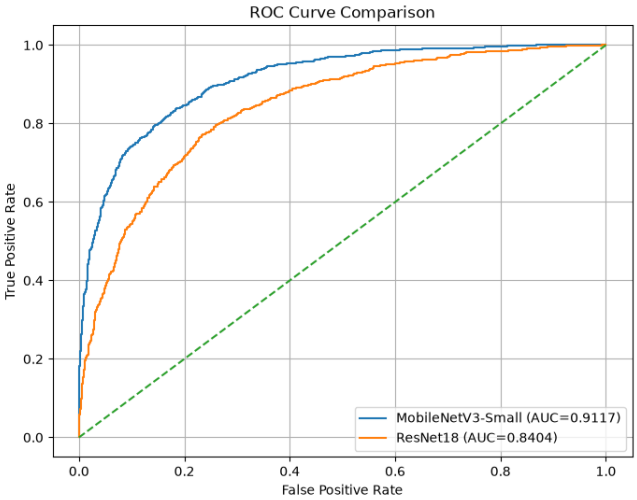


FIGURE 9. Initial-stage ROC comparison. These curves belong to the initial checkpoints, not the later fine-tuned or extended models.

for ResNet18. The accuracy difference is 5.30 percentage points. MobileNet has higher fake-class precision but lower fake recall than the initial ResNet, so the accuracy advantage does not mean it detects a larger proportion of fake images at the default threshold.

The initial MobileNet model correctly labels 1,318 real and 1,147 fake images, giving 182 false positives and 353 false negatives. ResNet18 correctly labels 1,148 real and 1,158 fake images, with 352 false positives and 342 false negatives. Thus, most of MobileNet's initial accuracy advantage comes from correctly rejecting more real images, while its number of detected fake images is slightly lower.

ROC-AUC and thresholded precision describe different properties. MobileNet's initial AUC advantage indicates better aggregate score ordering on this test set, while its lower fake recall reflects the particular operating point at 0.5. These observations motivate retaining probabilities and confusion matrices alongside headline accuracy.

D. FINE-TUNED EVALUATION

With prior optimization, MobileNet has reached 83.13% accuracy, 0.8045 precision, 0.8753 recall, 0.8384 F1, and 0.9173 ROC-AUC. ResNet18 achieves 83.87% accuracy, 0.8307 precision, 0.8507 recall, 0.8406 F1, and 0.9198 ROC-AUC. ResNet's accuracy advantage is 0.73 percentage points, corresponding to 22 additional correct classifications among 3,000 images. Its F1 advantage is approximately 0.0021 and its AUC advantage approximately 0.0025.

The two error profiles differ despite similar aggregate performance. MobileNet identifies 37 more fake images than ResNet18, but incorrectly flags 59 more real images. Its higher fake recall is therefore accompanied by a higher false-positive rate: 21.27% versus 17.33%. Its fake false-negative rate is 12.47%, compared with 14.93% for ResNet18. Which error pattern is preferable depends on the cost of missing synthetic content relative to flagging authentic content.

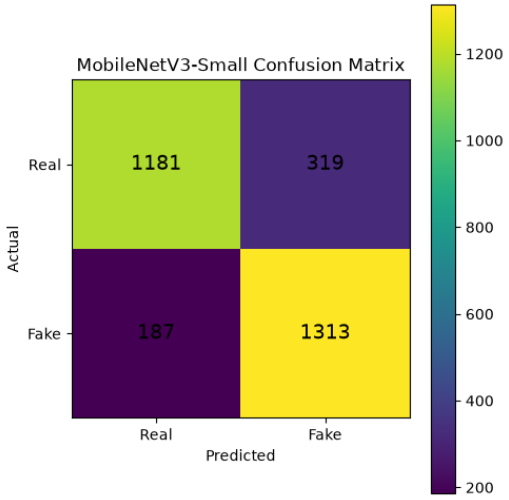


FIGURE 10. FIGURE 10. FIGURE 10 displays the confusion matrix, for the MobileNetV3-Small. We notice that the number of negatives is 1,181. We notice that the number of positives is 319. Also, the number of negatives is 187. Moreover, that the number of positives is 1,313.

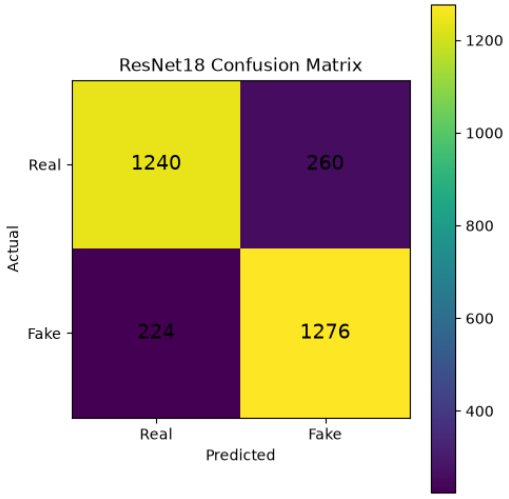


FIGURE 11. Figure 11 shows the confusion matrix, for the fine-tuned ResNet18. The number of negatives is 1,240. The number of positives is 260. The number of negatives is 224. The number of positives is 1,276.

Compared with initial training, MobileNet increases the number of correctly identified fake images from 1,147 to 1,313, while correct real classifications fall from 1,318 to 1,181. The net change is 29 additional correct images, or 0.97 percentage points in accuracy. ResNet18 improves both class recalls: correct real classifications increase by 92 and correct fake classifications by 118, giving 210 additional correct images and a 7.00-point accuracy improvement.

The larger improvement for ResNet18 reflects its weaker initial state and the particular fine-tuning procedure, rather than demonstrating that one architecture universally benefits more from adaptation. The comparison uses one recorded split and run. Without paired uncertainty analysis or repeated seeds, the small final differences remain descriptive point

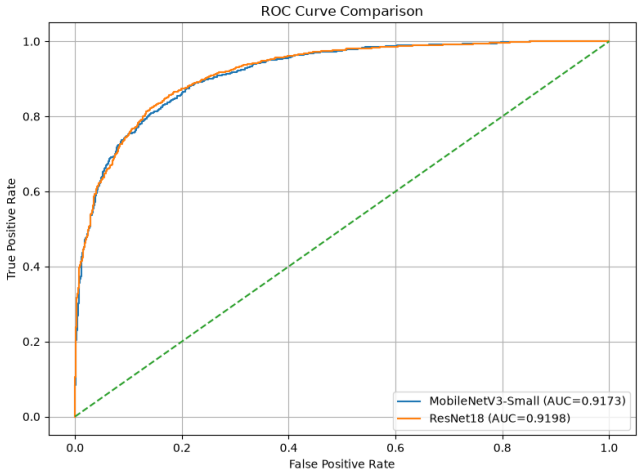


FIGURE 12. ROC curves for the earlier fine-tuned checkpoints. The close AUC values should not be interpreted as a statistically established tie or superiority.

TABLE 11. Extended MobileNet epochs. Epoch 10 is the lowest-validation-loss checkpoint. Epoch times are rounded notebook measurements.

Ep.	Train L	Train A	Val L	Val A	min
1	0.3257	0.8629	0.3249	0.8600	2.20
2	0.3219	0.8588	0.3130	0.8648	1.00
3	0.3094	0.8728	0.3056	0.8743	0.90
4	0.3106	0.8696	0.3036	0.8695	1.10
5	0.3058	0.8676	0.2985	0.8819	0.90
6	0.3004	0.8745	0.2948	0.8810	1.00
7	0.2940	0.8773	0.2876	0.8790	1.10
8	0.2932	0.8741	0.2863	0.8857	0.90
9	0.2828	0.8803	0.2882	0.8810	1.10
10	0.2725	0.8909	0.2786	0.8886	0.90

estimates.

E. EXTENDED MOBILENETV3-SMALL EVALUATION

The extended MobileNet run restores the saved earlier fine-tuned checkpoint and creates a new Adam optimizer with learning rate 0.00001 and weight decay 0.0001. It trains the final three feature blocks and classifier, corresponding to 1,236,513 trainable parameters out of 1,518,881 total parameters. Separate training and validation dataset instances preserve the same split while applying augmentation only to training images. Frozen earlier blocks are kept in evaluation mode.

Training loss drops from 0.3257 to 0.2725 while the validation loss drops from 0.3249 to 0.2786, over the training period. Training accuracy is increased from 86.29% to 89.09%, whereas validation accuracy is increased from 86.00% to 88.86%. The final training-validation accuracy difference is 0.23 percentage points, although augmented training and deterministic validation are not identical measurement conditions.

The validation loss generally decreases, with a temporary increase at epoch 9 from 0.2863 to 0.2882. The checkpoint is not replaced at that epoch. Epoch 10 yields the lowest

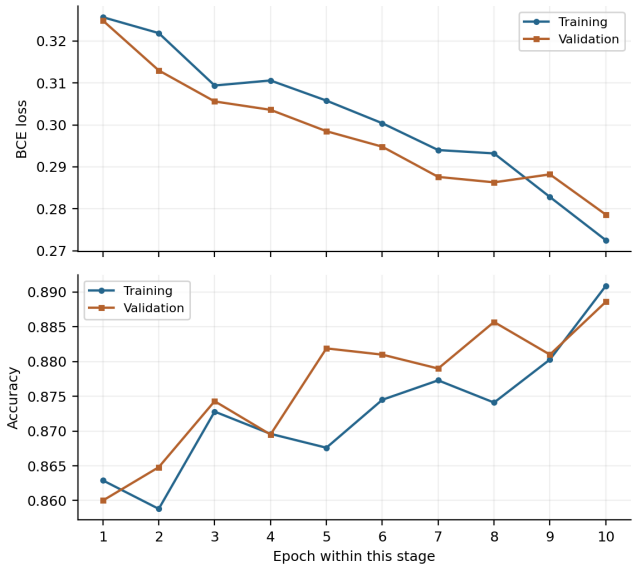


FIGURE 13. Extended MobileNet learning curves reconstructed from printed logs. Validation now uses deterministic preprocessing.

TABLE 12. Extended MobileNet clean test. The confusion-matrix counts support recomputing threshold metrics; AUC is retained from the saved output.

Metric	Value
Held-out images	3,000
Accuracy	0.8563 (85.63%)
Fake precision	0.8407
Fake recall	0.8793
Fake F1	0.8596
ROC-AUC	0.9355
Correct / incorrect	2,569 / 431

saved validation loss, reported more precisely as 0.27858, and becomes the selected model. Early stopping does not trigger before the ten-epoch cap. The sum of the rounded epoch times is approximately 11.1 minutes; this is training-session evidence, not a repeated runtime benchmark.

The extended model correctly labels 1,250 real images and 1,319 fake images. It incorrectly flags 250 real images and misses 181 fake images, giving real recall of 83.33% and fake recall of 87.93%. We noticed that the positive rate is 16.67% while the false-negative rate is 12.07%. These counts explain the reported 85.63% balanced-set accuracy and show that the model still produces substantial errors in both classes.

Relative to the earlier fine-tuned MobileNet, the extended checkpoint adds 75 correct classifications: 69 additional correct real classifications and six additional correct fake classifications. Accuracy rises by 2.50 percentage points, fake precision by approximately 0.0361, fake recall by 0.0040, F1 by 0.0211, and ROC-AUC by 0.0182. The reduction in false positives accounts for most of the improvement in thresholded accuracy.

This improvement is observed on the same 3,000-image directory already examined during earlier stages. The extended epoch is selected by internal validation loss, but

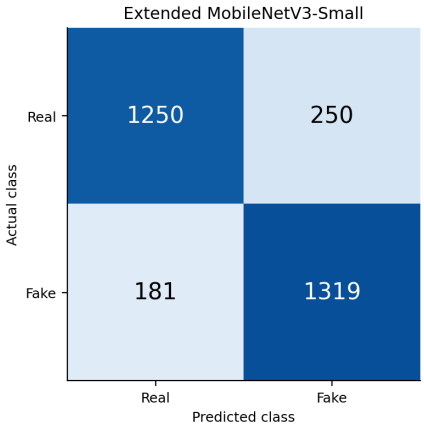


FIGURE 14. Extended MobileNet confusion matrix redrawn directly from the notebook's printed counts.

TABLE 13. Extended MobileNet class-specific metrics recomputed from its confusion matrix, shown to four decimals.

Class	Precision	Recall	F1	Support
Real	0.8735	0.8333	0.8530	1,500
Fake	0.8407	0.8793	0.8596	1,500

repeated inspection of the held-out set across development stages means the broader study is exploratory. A fresh external test set would provide a stronger independent assessment of the final development choices.

No extended MobileNet corruption sweep, Grad-CAM consistency rerun, or inference timing is saved. The earlier robustness results and CPU timing must therefore remain associated with the earlier fine-tuned checkpoint. Likewise, the notebook prints the extended AUC but does not embed its ROC curve or provide its raw probability vector in the experimental record; no extended ROC curve is fabricated here.

F. EXTENDED RESNET18 VALIDATION

The extended ResNet18 run restores the earlier fine-tuned checkpoint, freezes the early backbone, and trains `layer4` plus the final classifier. It reports 8,394,241 trainable parameters. As with extended MobileNet, validation is deterministic, checkpoint selection minimizes validation loss, Adam uses learning rate 0.00001 and weight decay 0.0001, and the maximum run length is ten additional epochs.

Loss for ResNet18 goes down from 0.3321 to 0.1385, while validation loss goes down from 0.3312 to 0.2389. Training accuracy is increased from 86.08% to 94.76%, while validation accuracy is increased from 85.62% to 91.24% The final training-validation gap is 3.52 percentage points. Both losses improve overall, but the larger reduction in training loss indicates a growing difference between training fit and validation performance.

The validation loss improves through epoch 8, increases slightly at epoch 9, and reaches its minimum at epoch 10.

TABLE 14. Extended ResNet18 epochs. These are training and internal validation results; they are not held-out test scores.

Ep.	Train L	Train A	Val L	Val A	min
1	0.3321	0.8608	0.3312	0.8562	3.90
2	0.2903	0.8802	0.3121	0.8695	4.30
3	0.2626	0.8923	0.2978	0.8781	3.70
4	0.2387	0.9087	0.2778	0.8848	4.20
5	0.2177	0.9151	0.2700	0.8867	4.30
6	0.1957	0.9261	0.2574	0.8971	4.10
7	0.1819	0.9343	0.2463	0.9000	4.30
8	0.1659	0.9375	0.2396	0.9048	3.60
9	0.1485	0.9477	0.2423	0.9029	3.60
10	0.1385	0.9476	0.2389	0.9124	3.60

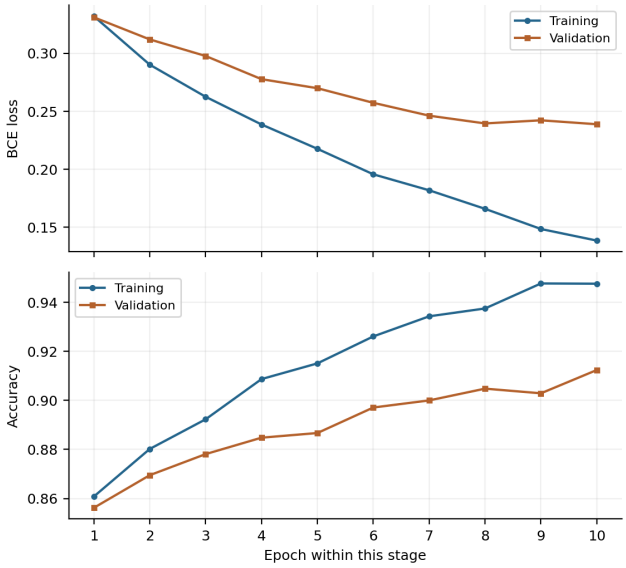


FIGURE 15. Extended ResNet18 learning curves reconstructed from its saved logs. No held-out curve or test result is available for this checkpoint.

The selected loss is reported as 0.23889. Early stopping is not reached within the ten-epoch limit. The rounded epoch times sum to approximately 39.6 minutes. The much longer run than extended MobileNet is consistent with a larger trainable model, but these session durations are not controlled measurements of computational complexity.

On internal validation at epoch 10, extended ResNet18 has 91.24% accuracy compared with 88.86% for extended MobileNet, a 2.38-point difference. ResNet18 also has lower validation loss. These are development-set comparisons under the extended protocol; they do not establish that extended ResNet18 has better held-out performance. In particular, 91.24% must not be reported as a test accuracy or compared directly with MobileNet's 85.63% test accuracy as though both were measured on the same split.

The saved notebook ends without evaluating the extended ResNet18 checkpoint on the image test set. Its test precision, recall, F1, AUC, and confusion matrix are therefore unavailable. The earlier ResNet18 result of 83.87% remains valid for the earlier checkpoint only. A complete updated architectural comparison requires a matching held-out evaluation of the extended ResNet18 state.

TABLE 15. MobileNetV3-Small: JPEG, n = 3,000.

Level	Acc.	Prec.	Recall	F1	AUC	s
90	0.8220	0.7875	0.8820	0.8321	0.9104	28.09
70	0.8023	0.7623	0.8787	0.8164	0.8861	29.13
50	0.7853	0.7533	0.8487	0.7981	0.8732	30.05
30	0.7703	0.7526	0.8053	0.7781	0.8512	30.99

TABLE 16. ResNet18: JPEG, n = 3,000.

Level	Acc.	Prec.	Recall	F1	AUC	s
90	0.8327	0.8244	0.8453	0.8348	0.9165	86.52
70	0.7947	0.8953	0.6673	0.7647	0.9085	85.83
50	0.7580	0.9339	0.5553	0.6965	0.8993	91.59
30	0.6637	0.9538	0.3440	0.5056	0.8791	90.02

G. ROBUSTNESS EVALUATION

The robustness record contains 19 corruption settings per model, giving 38 model-condition evaluations. JPEG quality 90, 70 and 50 and 30 is evaluated on all 3,000 held-out images. Gaussian blur radii 1 and 2 and 3 and 5; noise deviations 5 and 15 and 25 and 40; brightness factor 0.6 and 0.8 and 1.2 and 1.4; and scaled resolution 0.75 and 0.50 and 0.25 use the fixed stratified 500-image subset. Every subset condition therefore contains 250 real and 250 fake images.

Corruptions are applied to raw PIL images before resizing, tensor conversion, and normalization. The later noise evaluator uses a deterministic seed derived from image index and severity so both models see the same noisy realization. Noise standard deviation is expressed on the 0–255 pixel scale, with values clipped to that range. Resolution reduction uses bilinear downsampling followed by bilinear upsampling to the source dimensions before the normal model preprocessing.

The methodology specifies matched clean-subset baselines, but no clean result for the exact 500-image subset is saved. Consequently, clean-to-corrupt drops are calculated only for the full-set JPEG experiments. The remaining categories report absolute corrupted performance and changes between their evaluated severity settings. A brightness score above a full-set clean accuracy is not evidence that brightness improves detection, because the sample composition differs.

In the tables below, all five predictive metrics are included for every saved condition. The final column reports elapsed seconds from that condition's evaluation loop, including image processing. JPEG uses image-by-image inference, whereas the later subset evaluator batches 32 images. These times document execution but cannot be used as directly comparable latency measurements across corruption families. All figures in this robustness section are the notebook's final corrected plots, with explicit severity ordering.

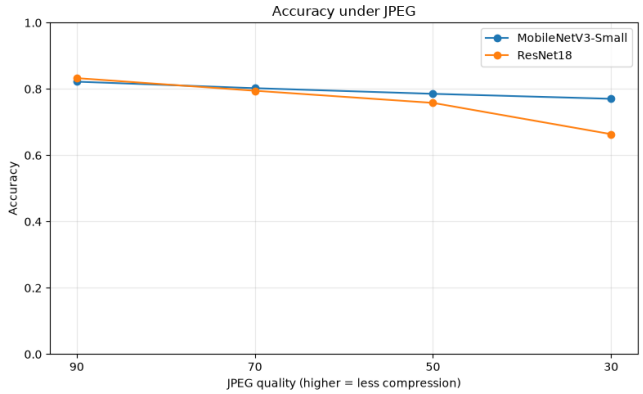


FIGURE 16. JPEG accuracy on all 3,000 held-out images. Lower quality indicates stronger compression; the checkpoints are the earlier fine-tuned models.

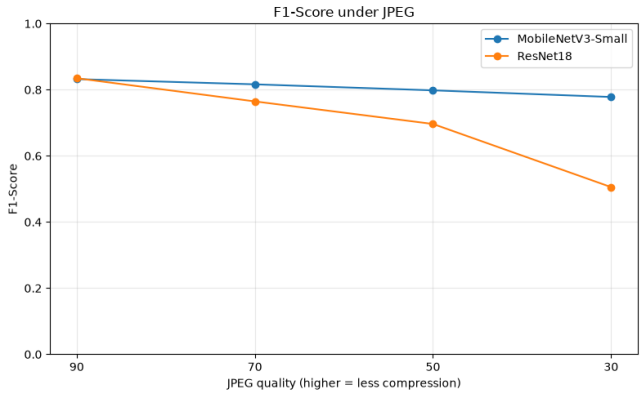


FIGURE 17. Fake-class F1 under JPEG compression. MobileNet retains higher F1 at qualities 70, 50, and 30.

H. JPEG COMPRESSION

At JPEG quality 90, performance remains close to the corresponding clean full-set results. MobileNet records 82.20% accuracy and 0.8321 F1, while ResNet18 records 83.27% and 0.8348. Relative to clean evaluation, the accuracy drops are 0.93 and 0.60 percentage points. At quality 70, MobileNet reaches 80.23% accuracy and ResNet18 79.47%; the ranking reverses as compression increases.

At quality 50, MobileNet retains 78.53% accuracy and 0.7981 F1, compared with 75.80% and 0.6965 for ResNet18. At quality 30, the difference becomes larger: MobileNet reaches 77.03% accuracy and 0.7781 F1, whereas ResNet18 reaches 66.37% and 0.5056. The strongest tested compression therefore produces a 10.67-point MobileNet accuracy advantage, or 320 additional correct classifications on the same images.

The precision-recall trajectory explains why ResNet18's compressed-image accuracy deteriorates. Its fake precision rises from 0.8244 at quality 90 to 0.9538 at quality 30, while recall falls from 0.8453 to 0.3440. At quality 30 it detects only 516 of 1,500 fake images, missing 984. Its high precision reflects a small set of mostly correct fake predictions, rather than reliable coverage of the fake class.

TABLE 17. JPEG accuracy loss relative to each model's own clean full-set score. Values are percentage-point decreases.

Quality	Mobile drop	ResNet drop
90	0.93	0.60
70	2.90	4.40
50	4.60	8.07
30	6.10	17.50

TABLE 18. MobileNetV3-Small: Gaussian Blur, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
1	0.8220	0.7949	0.8680	0.8298	0.8965	4.79
2	0.5460	0.8286	0.1160	0.2035	0.7880	3.48
3	0.5140	1.0000	0.0280	0.0545	0.7101	3.45
5	0.5040	1.0000	0.0080	0.0159	0.6484	3.97

TABLE 19. ResNet18: Gaussian Blur, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
1	0.8160	0.8873	0.7240	0.7974	0.9055	10.94
2	0.6180	0.9275	0.2560	0.4013	0.8470	11.54
3	0.4980	0.0000	0.0000	0.0000	0.7636	12.55
5	0.5000	0.0000	0.0000	0.0000	0.6151	12.30

MobileNet detects 1,208 fake images at quality 30, missing 292. From the reported recall and accuracy, the corresponding confusion counts are TN $=$ 1,103, FP $=$ 397, FN $=$ 292, and TP $=$ 1,208. ResNet18 has TN $=$ 1,475, FP $=$ 25, FN $=$ 984, and TP $=$ 516. These counts, derived from the full-set metrics, make the trade-off explicit: MobileNet catches many more fake images while generating more false alarms.

ResNet18 still has a higher quality-30 AUC, 0.8791 versus 0.8512, despite much worse thresholded accuracy and F1. This suggests that score ordering retains information while the default threshold yields poor fake recall under compression. It does not establish that a different threshold would solve the problem. Any threshold adaptation would need development data and a separate evaluation, rather than tuning on these reported test conditions.

I. GAUSSIAN BLUR

At blur radius 1, the two earlier checkpoints maintain similar subset accuracies: 82.20% for MobileNet and 81.60% for ResNet18. MobileNet has higher fake recall, 0.8680 versus 0.7240, whereas ResNet18 has higher precision, 0.8873 versus 0.7949. A mild blur therefore already produces different operating-point balances even when aggregate accuracies are close.

Increasing the radius from 1 to 2 reduces MobileNet accuracy by 27.60 percentage points, to 54.60%, and ResNet18 accuracy by 19.80 points, to 61.80%. The fake recalls at radius 2 are only 0.1160 and 0.2560. This means MobileNet detects 29 of 250 fake images and ResNet18 detects 64. The decline is therefore dominated by fake images being assigned

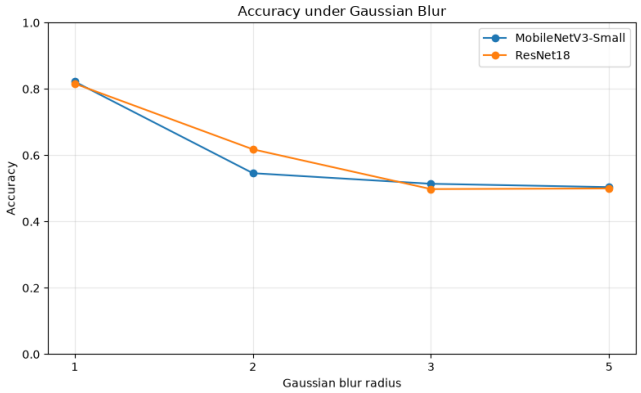


FIGURE 18. Accuracy under Gaussian blur on the same balanced 500-image subset. Both curves approach 0.5 at the stronger settings.

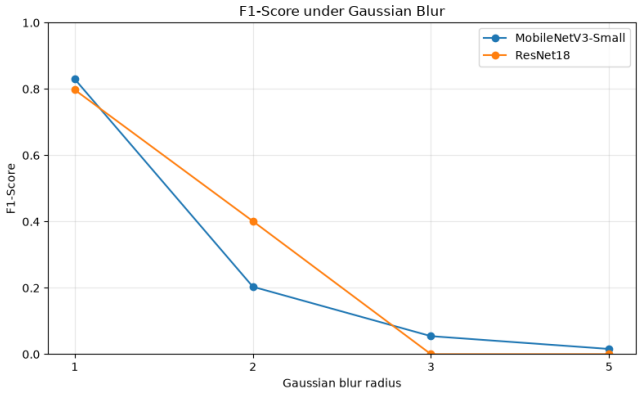


FIGURE 19. Fake-class F1 under blur. Near-zero F1 reveals the failure to detect fake images that accuracy alone can obscure.

to the real class.

At radius 3, MobileNet reaches 51.40% accuracy, precision 1.0000, recall 0.0280, and F1 0.0545. Perfect precision occurs because the model flags only seven fake images and no real images; it misses the other 243 fake images. At radius 5, it flags only two fake images and misses 248, yielding accuracy 50.40% and F1 0.0159. Precision must be interpreted with recall and predicted-positive count.

ResNet18 detects no fake images at either radius 3 or 5. At radius 3, accuracy 49.80% corresponds to 249 correctly accepted real images, one real false positive, and 250 missed fake images. At radius 5, accuracy is exactly 50.00% and the model predicts every image as real. Thus the apparent chance-level accuracy reflects a strongly biased decision rule, not evidence that the model is making balanced random errors.

AUC remains above 0.5 for these severe blur conditions, including 0.7636 for ResNet18 at radius 3, so continuous score ordering and the default decision threshold behave differently. Nonetheless, the very low fake recall makes the tested operating point unsuitable for detecting blurred fake images. Sensitivity to removed spatial detail is a plausible interpretation, but the experiment does not isolate particular frequency components or prove which features caused the failure.

TABLE 20. MobileNetV3-Small: Gaussian Noise, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
5	0.7900	0.7245	0.9360	0.8168	0.9099	4.54
15	0.6920	0.6244	0.9640	0.7579	0.8610	4.94
25	0.6280	0.5777	0.9520	0.7190	0.7822	4.39
40	0.5800	0.5474	0.9240	0.6875	0.7037	5.26

TABLE 21. ResNet18: Gaussian Noise, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
5	0.8120	0.7583	0.9160	0.8297	0.9142	13.36
15	0.7980	0.7651	0.8600	0.8098	0.8720	12.89
25	0.7440	0.7542	0.7240	0.7388	0.8128	13.10
40	0.6400	0.6750	0.5400	0.6000	0.6942	13.55

J. GAUSSIAN NOISE

Noise reduces accuracy for both models as standard deviation increases. At sigma 5, MobileNet has 79.00% accuracy and ResNet18 81.20%. At sigma 15 the results are 69.20% and 79.80%; at sigma 25 they are 62.80% and 74.40%; and at sigma 40 they are 58.00% and 64.00%. ResNet18 has higher accuracy at all four measured noise levels, with its largest advantage, 11.60 percentage points, at sigma 25.

The models differ in how they fail. MobileNet recall remains high, between 0.9240 and 0.9640 at the three stronger levels, while precision decreases from 0.7245 at sigma 5 to 0.5474 at sigma 40. At sigma 40, 231 of the 250 fake images are detected, but 191 real images are incorrectly flagged. The resulting confusion counts are TN $= 59$, FP $= 191$, FN $= 19$, and TP $= 231$. The high recall is obtained at a false-positive rate of 76.40%.

ResNet18 at sigma 40 has precision 0.6750 and recall 0.5400, corresponding to TN $= 185$, FP $= 65$, FN $= 115$, and TP $= 135$. It is more conservative about fake predictions and preserves more real-image correctness, giving higher overall accuracy. However, its fake F1 is 0.6000 versus MobileNet's 0.6875. Accuracy and positive-class F1 can therefore rank the same two models differently even on a balanced dataset.

The noise AUC values also decline with severity. At sigma 40, MobileNet records 0.7037 and ResNet18 0.6942. The small reversed AUC ordering at this setting reinforces the distinction between score discrimination and thresholded class decisions. The results support a condition-specific trade-off rather than a single universal robustness winner.

Because the noise realization is fixed per image and severity, the paired architecture comparison uses controlled inputs. It still represents only one realization per setting and one sampled subset. Repeated noise draws and additional subsets would be required to estimate variation attributable to the corruption process and sample selection.

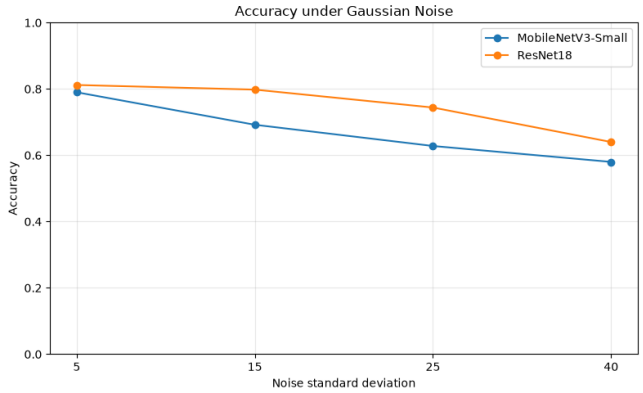


FIGURE 20. Accuracy under deterministic additive Gaussian noise, measured on the fixed 500-image subset. Sigma is defined on the 0–255 pixel scale.

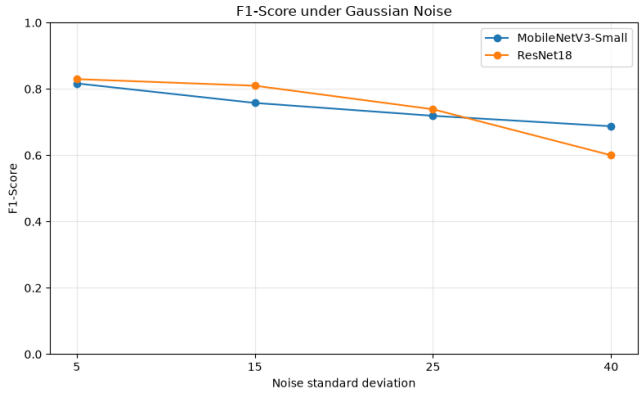


FIGURE 21. Fake-class F1 under Gaussian noise. At sigma 40, the model with higher accuracy has lower fake-class F1.

TABLE 22. MobileNetV3-Small: Brightness, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
0.6	0.8300	0.8287	0.8320	0.8303	0.9140	2.73
0.8	0.8580	0.8510	0.8680	0.8594	0.9283	2.74
1.2	0.8560	0.8477	0.8680	0.8577	0.9272	2.75
1.4	0.8280	0.8333	0.8200	0.8266	0.9090	2.76

TABLE 23. ResNet18: Brightness, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
0.6	0.8480	0.8452	0.8520	0.8486	0.9289	11.69
0.8	0.8480	0.8321	0.8720	0.8516	0.9321	10.82
1.2	0.8540	0.8831	0.8160	0.8482	0.9239	10.83
1.4	0.8020	0.8832	0.6960	0.7785	0.9024	10.74

K. BRIGHTNESS

Brightness is the least disruptive of the five tested categories over the selected factors. MobileNet accuracy ranges from 82.80% to 85.80%; ResNet18 ranges from 80.20% to 85.40%. These factors describe both darkening and brightening around the untested identity setting of 1.0. They should not be interpreted as a monotonic severity scale from left to right.

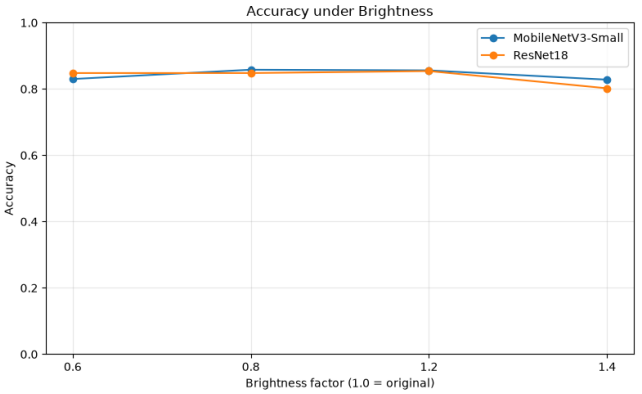


FIGURE 22. Accuracy under brightness factors 0.6, 0.8, 1.2, and 1.4 on the fixed subset. Factor 1.0 is not included in the saved experiment.

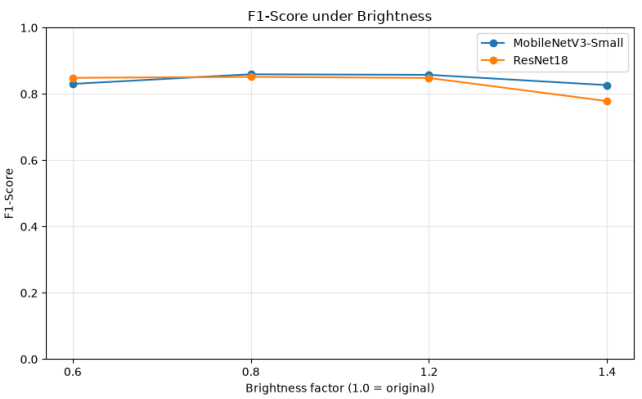


FIGURE 23. Fake-class F1 under brightness changes. Both darkening and brightening are represented; no matched clean-subset result is available.

At factor 0.6, ResNet18 has 84.80% accuracy compared with MobileNet's 83.00%. At factor 0.8, MobileNet reaches 85.80% and ResNet18 84.80%. At factor 1.2 the models are close, at 85.60% and 85.40%. At factor 1.4, MobileNet retains 82.80% while ResNet18 falls to 80.20%. The model ordering changes across factors, so brightness robustness is not summarized adequately by one selected condition.

Under the strongest tested brightening, MobileNet precision and recall are 0.8333 and 0.8200, while ResNet18 has precision 0.8832 and recall 0.6960. MobileNet detects 205 fake images and ResNet18 detects 174. ResNet18 produces fewer real false positives, 23 versus 41, but misses more fake images, 76 versus 45. This is a milder version of the conservative fake-decision pattern also observed under compression and resolution loss.

The observed stability is consistent with the training pipeline including color jitter, but no augmentation ablation is present. The experiment therefore cannot establish that color jitter caused brightness tolerance. Similarly, the subset results of 85.80% and 85.40% must not be described as clean-score improvements over the full 3,000-image evaluation. A factor-1.0 run on these exact 500 images is required for that comparison.

TABLE 24. MobileNetV3-Small: Resolution, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
0.75	0.8260	0.7801	0.9080	0.8392	0.9075	3.27
0.5	0.7640	0.8367	0.6560	0.7354	0.8660	3.14
0.25	0.6260	0.9091	0.2800	0.4281	0.8063	3.07

TABLE 25. ResNet18: Resolution, n = 500.

Level	Acc.	Prec.	Recall	F1	AUC	s
0.75	0.8280	0.8868	0.7520	0.8139	0.9160	11.29
0.5	0.7620	0.9226	0.5720	0.7062	0.8932	11.09
0.25	0.5220	0.9231	0.0480	0.0913	0.8327	10.96

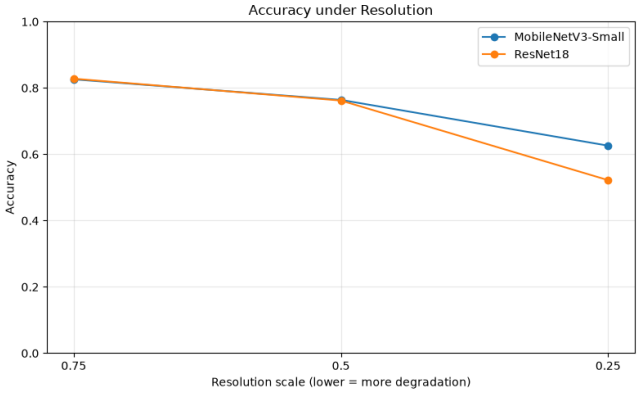


FIGURE 24. Accuracy under resolution reduction and upscaling. Lower scale means greater spatial information loss; n = 500 for each point.

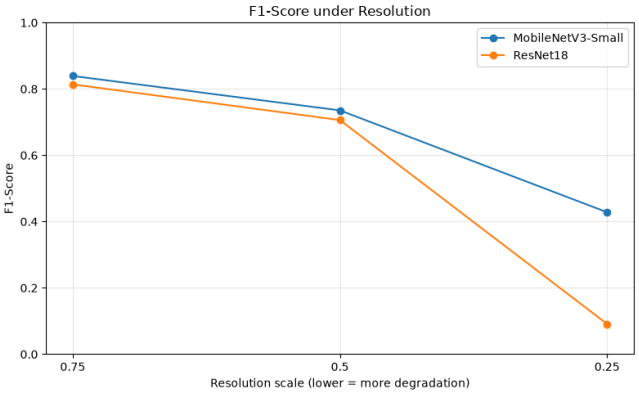


FIGURE 25. Fake-class F1 under resolution reduction. The sharp drop at scale 0.25 is especially pronounced for ResNet18.

TABLE 26. Unweighted means across the tested severities within each category, recomputed from saved table values. M = MobileNet; R = ResNet.

Category	Model	Mean A	Mean F1	Mean AUC
JPEG	M	0.7950	0.8062	0.8802
JPEG	R	0.7623	0.7004	0.9008
Gaussian Blur	M	0.5965	0.2759	0.7607
Gaussian Blur	R	0.6080	0.2997	0.7828
Gaussian Noise	M	0.6725	0.7453	0.8142
Gaussian Noise	R	0.7485	0.7446	0.8233
Brightness	M	0.8430	0.8435	0.9197
Brightness	R	0.8380	0.8317	0.9218
Resolution	M	0.7387	0.6676	0.8599
Resolution	R	0.7040	0.5371	0.8806

L. RESOLUTION

Resolution tests reduce each source image to a fraction of its original width and height, then upscale it before standard preprocessing. The severity parameter therefore refers to source-image dimensions, not necessarily a fixed intermediate size such as 56 by 56 pixels. The notebook does not print all original dimensions, so no universal reduced pixel size is assumed.

At scale 0.75, MobileNet and ResNet18 have similar accuracies of 82.60% and 82.80%, respectively. Their fake recalls differ substantially, however: 0.9080 for MobileNet and 0.7520 for ResNet18. At scale 0.50, accuracy remains closely matched at 76.40% and 76.20%, while MobileNet retains higher fake recall, 0.6560 versus 0.5720, and higher F1, 0.7354 versus 0.7062.

At scale 0.25, MobileNet accuracy is 62.60% and ResNet18 accuracy is 52.20%, a 10.40-point difference corresponding to 52 additional correct subset predictions. MobileNet's fake F1 is 0.4281, compared with 0.0913 for ResNet18. Their recalls are 0.2800 and 0.0480, meaning that only 70 and 12 of 250 fake images are detected.

The quarter-scale confusion counts derived from these metrics are TN = 243, FP = 7, FN = 180, TP = 70 for MobileNet and TN = 249, FP = 1, FN = 238, TP = 12 for ResNet18. Both models predominantly predict real under severe resolution loss. High fake precision, 0.9091 and 0.9231, does not compensate for the large fraction of undetected fake images.

ResNet18 retains a higher quarter-scale AUC, 0.8327 versus 0.8063, despite its worse default-threshold performance. As with JPEG, this suggests an operating-point mismatch alongside degradation in the available information. It is not evidence that a deployment-ready correction has been demonstrated. Any proposed recalibration or degradation-aware threshold would require separate training or validation and further held-out evaluation.

M. CORRUPTION SUMMARY

Averaging within a corruption family gives MobileNet higher mean accuracy for JPEG, brightness, and resolution, and ResNet18 higher mean accuracy for blur and noise. The blur mean should be interpreted carefully: both models fail almost completely to detect fake images at the strongest radii. A small average advantage does not make either model robust across that category.

These means are descriptive summaries of the chosen grid. They are not standardized benchmark scores, do not represent a natural frequency distribution of image distortions, and do not combine clean and corrupted samples into a single test. Different numbers of levels and different sample sizes across categories make an overall average across all conditions

TABLE 27. Full min-max summaries across each tested category. Minima in different metric columns need not identify the same condition.

Category	M/R	Min A	Max A	Min F1	Max F1
JPEG	M	0.7703	0.8220	0.7781	0.8321
JPEG	R	0.6637	0.8327	0.5056	0.8348
Gaussian Blur	M	0.5040	0.8220	0.0159	0.8298
Gaussian Blur	R	0.4980	0.8160	0.0000	0.7974
Gaussian Noise	M	0.5800	0.7900	0.6875	0.8168
Gaussian Noise	R	0.6400	0.8120	0.6000	0.8297
Brightness	M	0.8280	0.8580	0.8266	0.8594
Brightness	R	0.8020	0.8540	0.7785	0.8516
Resolution	M	0.6260	0.8260	0.4281	0.8392
Resolution	R	0.5220	0.8280	0.0913	0.8139

TABLE 28. Model size and earlier fine-tuned clean evaluation timing. Elapsed time includes the evaluation loop and its data loading / preprocessing.

Quantity	MobileNet	ResNet18
Total parameters	1,518,881	11,177,025
Clean elapsed seconds	16.2842	72.5933
Images per second	184.23	41.33
Amortized ms/image	5.43	24.20
Extended trainable params	1,236,513	8,394,241

difficult to interpret. No such average is used to declare a universal winner.

Three distinct observed failure patterns emerge. Blur and strong downsampling largely suppress fake detections, producing many false negatives. Heavy noise makes MobileNet produce excessive fake predictions, creating many false positives. JPEG compression causes ResNet18's fake recall to fall steeply while precision remains high. A single accuracy number hides these operationally different outcomes.

On a balanced 500-image subset, one changed classification alters accuracy by 0.20 percentage points. On the full 3,000-image set, one classification changes accuracy by approximately 0.033 percentage points. Small numerical differences should therefore be read in relation to the sample size, and six printed decimal places should not be mistaken for high statistical certainty. No confidence intervals or paired significance tests are saved in the notebook.

N. COMPUTATIONAL EFFICIENCY

MobileNetV3-Small contains approximately 1.52 million parameters, compared with 11.18 million for ResNet18. This is an 86.41% reduction in parameter count, or a ResNet-to-MobileNet ratio of approximately 7.36. The count describes learned parameter elements, not serialized checkpoint size, activation memory, optimizer state, or peak runtime memory.

The earlier fine-tuned clean evaluation takes 16.2842 seconds for MobileNet and 72.5933 seconds for ResNet18 across 3,000 images. Reported throughput is 184.23 and 41.33 images per second, respectively, a ratio of approximately 4.46 in this recorded run. Dividing total time by image count gives amortized processing costs of 5.43 and 24.20 milliseconds per image. These averages are batch-evaluation quantities, not measured single-image request latency.

TABLE 29. Initial-stage clean timing also retained in the notebook; a different model stage and measurement run.

Quantity	MobileNet	ResNet18
Elapsed seconds	16.844543	66.581617
Images per second	178.099220	45.057482

The timer starts before iteration over the data loader. With zero loader workers, image loading and preprocessing occur inside the measured loop, alongside model inference and result collection. It is therefore incorrect to describe these measurements as forward-pass-only latency or to claim that preprocessing is excluded. The code also does not implement an explicit warm-up and repeated-timing study for the reported clean comparison.

The initial and earlier fine-tuned timing results differ even though architecture parameter counts are unchanged. This variation reinforces that a single observed timing reflects the session and evaluation conditions. The extended training cells explicitly set four CPU threads, but no comparable extended inference timing is saved. Neither initial nor earlier fine-tuned timings are relabeled as an extended-checkpoint benchmark.

Corruption-loop times are included with every corruption table for completeness. JPEG times include compression and single-image inference over 3,000 images, whereas later subset times include image corruption and batched inference over 500 images. Dividing or comparing these totals without accounting for sample count, batch policy, and preprocessing would conflate several effects. They provide execution context, not evidence that a corruption changes intrinsic architecture speed.

The results support MobileNet as the smaller and faster observed option under the reported CPU protocol. They do not establish energy efficiency, mobile-device latency, real-time video throughput, or deployment memory consumption. Such claims would require measurements on the intended hardware and workload.

O. GRAD-CAM

Grad-CAM is generated at the final MobileNet feature block and the last block of ResNet18's `layer4`. The binary target uses the output logit for fake and its negative for real. Unless explicitly fixed, the target is the predicted class. This convention matters because a heatmap for a mistaken real prediction describes support for that prediction, not necessarily support for the ground-truth fake class.

The four-image comparison uses source indices 0, 100, 500, and 1000. All four displayed examples belong to the fake class, as confirmed by the figure labels. MobileNet predicts fake for each, with displayed probabilities 0.974, 0.921, 0.896, and 0.842. ResNet18 predicts fake for three, with probabilities 0.961, 0.949, and 0.987, but predicts real for index 500 with $P(\text{fake}) = 0.397$.

The overlays emphasize different regions of the same

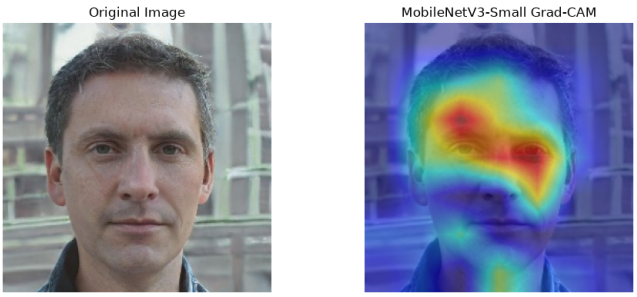


FIGURE 26. Single-image MobileNet Grad-CAM example. The saved output gives P(fake) = 0.9744 and a 224 by 224 heatmap.

TABLE 30. Explainability similarity for 20-images per model. SD = Sample Standard Deviation. It is not an uncertainty interval for the mean.

Statistic	MobileNet	ResNet18
Pairs	20	20
Mean	0.930226	0.901721
SD	0.058124	0.149931
Minimum	0.728745	0.472825
Maximum	0.995672	0.990292

TABLE 31. MobileNetV3-Small: 20 explanation records at JPEG quality 50. Pred = clean prediction (0 real, 1 fake); cosine compares maps targeting that same class.

Index	P(fake)	Pred	Cosine
256	0.773457	1	0.965566
266	0.307894	0	0.915544
281	0.984840	1	0.978217
384	0.886993	1	0.728745
602	0.893711	1	0.995672
1292	0.801049	1	0.951473
1309	0.912165	1	0.983199
1351	0.125074	0	0.955886
1538	0.718547	1	0.914255
1574	0.151260	0	0.953319
1952	0.447967	0	0.901255
2083	0.890068	1	0.948777
2148	0.434859	0	0.893274
2202	0.223435	0	0.914234
2278	0.027424	0	0.975979
2307	0.075200	0	0.955201
2355	0.684945	1	0.919619
2518	0.139652	0	0.901128
2563	0.405799	0	0.973138
2919	0.051253	0	0.880050

images. In the third example, MobileNet highlights facial regions including the eyes and mouth, whereas ResNet18's real-class explanation includes a strong response near the upper-right hair accessory and the lower-right image region. This visual difference is useful for identifying a case to investigate, but it does not establish the causal feature responsible for the error.

Because the targets are each model's own prediction, the third row compares a fake-target MobileNet explanation with a real-target ResNet18 explanation. The other displayed rows compare fake-target explanations. This distinction prevents treating every pair of overlays as a controlled same-target comparison. Furthermore, four selected fake images cannot establish explanation quality on authentic images or the prevalence of any attention pattern.

P. EXPLANATION CONSISTENCY

The explanation-consistency study samples 20 image indices without replacement using seed 42 and evaluates both models on the same clean/JPEG-quality-50 pairs. For each model and image, the clean predicted class is used as the target for both heatmaps. Cosine similarity is then computed between flattened maps. Holding the target fixed avoids confounding a target-class change with a change in spatial explanation.

MobileNet has higher mean similarity by approximately 0.0285 and a smaller sample standard deviation. The lowest similarity value is 0.7287 for MobileNet and 0.4728 for ResNet18. ResNet18 has two very low values. Index 384 has a similarity of 0.4917 and index 2563 has a similarity of 0.4728. Many of its other similarities are above 0.95. Including the full similarity values provides more information.

MobileNet does not show more consistency for all the sampled images. Index 2355 has a higher similarity for ResNet (0.9903 vs 0.9196). Index 2518 also has a higher similarity for ResNet (0.9810 vs 0.9011). The mean has been calculated using this sample of paired images. No claims can be made about statistical superiority.

The saved consistency table records the clean probability, clean predicted class, and heatmap similarity. The consistency table does not include the compressed probability, prediction or ground-truth labels. This means nothing can be said about how often the predictions change during JPEG or if consistent explanations relate to correct decisions. A high cosine similarity just means the two flattened maps have a similar spatial pattern during this process.

If a map has a zero norm the similarity is returned as 0. None of the similarity values have been zero but this should be mentioned. The similarity will depend on the chosen layer, spatial resolution and map. This does not prove the explanations are faithful, identify the causal region of the image or that the detector remains accurate after compression.

Q. IMAGE-LEVEL EXPLANATIONS

The target class may not always be the same for the two models. Each model uses its own clean prediction. Index 266 has MobileNet predicting real and ResNet18 predicting fake. The cosine similarity has been matched internally but the target class will be different between models. This is another reason why the mean values should be viewed as stability under the implemented rule rather than as a controlled ranking of explanation quality.

Index 384 has both models predicting fake for the clean image but has a lower similarity. MobileNet has a similarity of 0.728745 and ResNet18 has a similarity of 0.491706. Index 2563 has both models predicting different classes for the clean image. MobileNet has a similarity of 0.973138 and ResNet18 has a similarity of 0.472825. This shows why it is

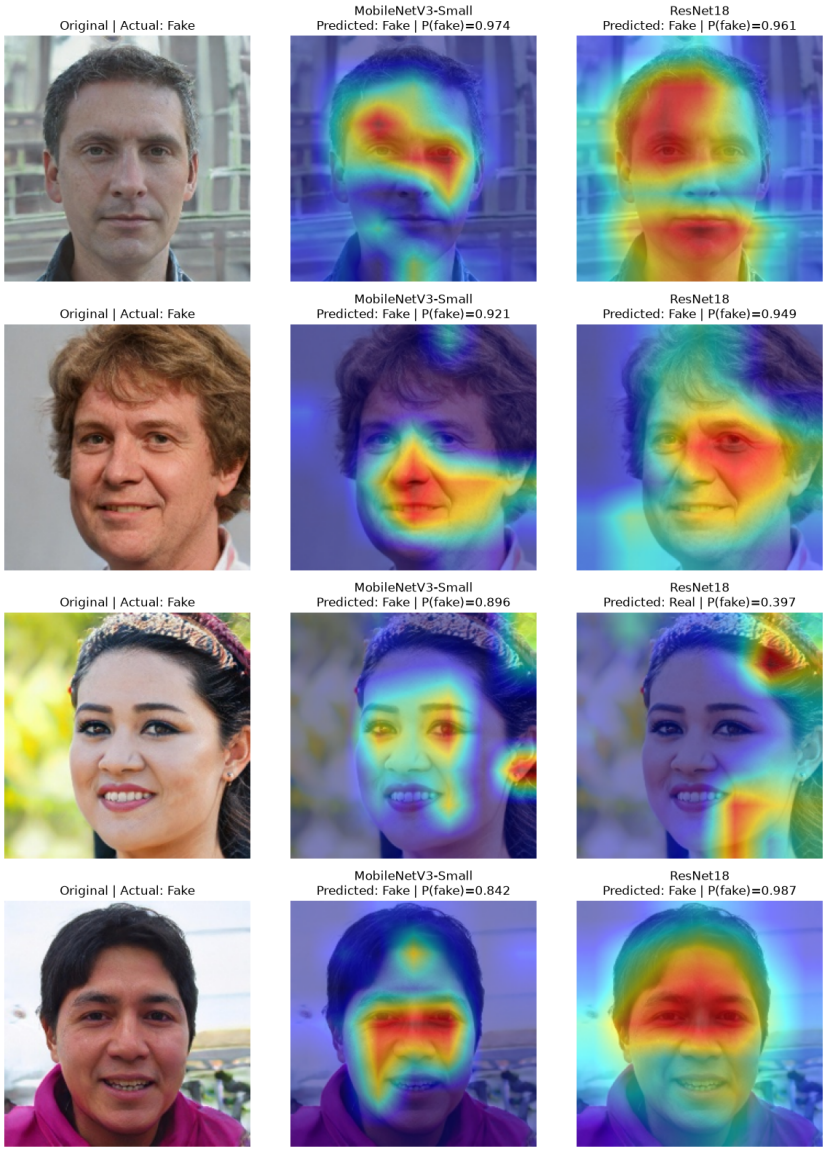


FIGURE 27. Four-image Grad-CAM comparison from the notebook. Rows correspond to indices 0, 100, 500, and 1000; columns show the source image, MobileNet overlay, and ResNet18 overlay. All four ground-truth labels are fake. The third-row ResNet explanation targets its predicted real class.

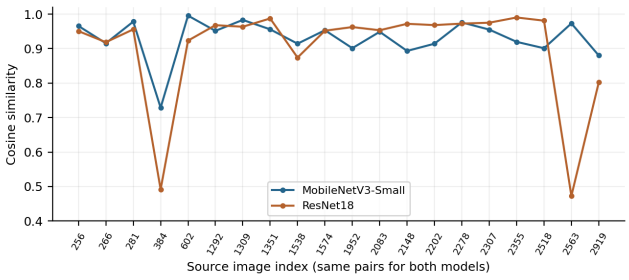


FIGURE 28. Heatmap cosine similarities for all 20 paired source indices, reconstructed from the saved 40-row table. Each line represents one model on the same image pairs.

important to include both stable and unstable examples in the representative visualisation.

All values in these two tables are retained to the six decimal places displayed by the notebook. They are observations from the earlier fine-tuned checkpoints and are not recomputed explanations for the extended models. Repeated appearances of this table elsewhere in the notebook represent redisplays of the same results and are not additional samples.

V. DISCUSSION

A. RQ1: CLASSIFICATION PERFORMANCE

The comparable earlier fine-tuned checkpoints achieve 83.13% accuracy for MobileNetV3-Small and 83.87% for ResNet18 on the same 3,000 images. ResNet18's advantage is 22 correct classifications, or approximately 0.73 percentage points, with a small fake-class F1 advantage (0.8406

TABLE 32. ResNet18: 20 explanation records at JPEG quality 50. Pred = clean prediction (0 real, 1 fake); cosine compares maps targeting that same class.

Index	P(fake)	Pred	Cosine
256	0.980543	1	0.950665
266	0.760566	1	0.918574
281	0.786365	1	0.955794
384	0.642446	1	0.491706
602	0.950337	1	0.923157
1292	0.129898	0	0.967945
1309	0.261249	0	0.963421
1351	0.055206	0	0.987394
1538	0.562913	1	0.874040
1574	0.286309	0	0.951947
1952	0.296972	0	0.962475
2083	0.760514	1	0.953253
2148	0.019126	0	0.971943
2202	0.089141	0	0.968180
2278	0.012664	0	0.972542
2307	0.028938	0	0.975127
2355	0.149212	0	0.990292
2518	0.069466	0	0.981041
2563	0.645641	1	0.472825
2919	0.356837	0	0.802103

versus 0.8384) and ROC-AUC advantage (0.9198 versus 0.9173). The error profiles are more informative than this narrow aggregate difference: MobileNet detects 37 additional fake images but incorrectly flags 59 additional real images. Its fake recall is 0.8753, compared with 0.8507 for ResNet18, while its precision is lower (0.8045 versus 0.8307). Thus, the two models offer different balances between missed synthetic content and false alarms. No repeated-seed or paired significance analysis establishes that the small clean-performance difference is statistically reliable. Precision measured on this balanced dataset also need not transfer to a workload with a different prevalence of fake content.

B. RQ2: CORRUPTION ROBUSTNESS

Robustness depends on the corruption and operating point. At JPEG quality 30, MobileNet retains 77.03% accuracy and 0.7781 fake-class F1, compared with 66.37% and 0.5056 for ResNet18; at resolution scale 0.25, the corresponding accuracies are 62.60% and 52.20%. ResNet18 instead has higher accuracy at every tested Gaussian-noise level, including 64.00% versus 58.00% at noise standard deviation 40. Both networks fail severely under strong blur: at radius 5, MobileNet detects only two of 250 fake images and ResNet18 detects none. Brightness changes are comparatively less disruptive over the evaluated factors. These findings identify different error mechanisms at the prediction level: blur and downsampling mainly increase false negatives, whereas heavy noise produces many false positives for MobileNet. Higher ROC-AUC can coexist with poor recall at threshold 0.5, but this does not establish that recalibration will resolve the failure. Clean-relative degradation is supported for full-set JPEG evaluation only; the absent matched clean baseline for the 500-image subset prevents equivalent degradation claims for the other four families.

C. RQ3: COMPUTATIONAL EFFICIENCY

MobileNetV3-Small provides the more compact measured option: 1,518,881 parameters versus 11,177,025 for ResNet18, an 86.41% reduction. For the earlier fine-tuned clean evaluation, it processes 3,000 images in 16.2842 seconds, compared with 72.5933 seconds for ResNet18, corresponding to approximately 184.23 and 41.33 images per second. This is a throughput ratio of about 4.46 in the recorded CPU loop, alongside a clean-accuracy difference of only 0.73 percentage points. The result supports an accuracy-efficiency trade-off on this workload. The timer includes data loading, preprocessing, inference, and result collection, however, and no explicit warm-up or repeated-run dispersion is available. The amortized costs of 5.43 and 24.20 milliseconds per image must therefore not be interpreted as isolated single-request latency. Parameter count and this one execution record do not establish mobile-device performance, energy consumption, memory usage, or real-time video capability.

D. RQ4: EXTENDED FINE-TUNING

Extended MobileNet fine-tuning raises held-out accuracy from 83.13% to 85.63%, with fake-class F1 of 0.8596 and ROC-AUC of 0.9355. The improvement comprises 75 additional correct classifications: 69 additional real images and six additional fake images. It is therefore driven mainly by fewer false positives. The extended procedure also changes validation preprocessing, checkpoint selection, and frozen-module handling, so the gain cannot be attributed uniquely to extra epochs. Extended ResNet18 reaches 91.24% internal validation accuracy, but no corresponding held-out evaluation is present; this development result cannot be compared with MobileNet's 85.63% test accuracy as an architectural ranking. The extended MobileNet checkpoint also lacks a repeated corruption sweep, explanation study, and inference timing. Earlier results consequently remain associated with their original checkpoints. Repeated inspection of the same held-out directory makes the broader study exploratory; an independent external evaluation would strengthen the final comparison.

E. RQ5: EXPLANATION CONSISTENCY

Grad-CAM reveals differences in the regions associated with individual predictions, including an incorrectly classified fake image whose ResNet18 overlay targets the predicted real class. Across 20 clean/JPEG-quality-50 pairs per model, MobileNet has mean heatmap cosine similarity 0.930226 with sample standard deviation 0.058124, compared with 0.901721 and 0.149931 for ResNet18. This supports greater average spatial consistency for MobileNet in the selected sample. The results do not show a universal ordering of the images. ResNet18 has higher heatmap similarity on some images. Both heatmaps are targeting the clean predicted class for the model-image pair but could vary between models. All four qualitative examples are fake images and the consistency table does not contain any compressed predictions or ground-truth labels. This means nothing can be said about how often

the predictions change, how accurate the localisation is or if the explanations are faithful. Having fixed targets, both classes, correct and incorrect examples and checking for perturbations/randomisations would provide a better validation of the explanations [29].

VI. LIMITATIONS

A. IMPLEMENTATION

The original methodology describes deterministic validation, minimum-validation-loss selection, matching clean subset baselines and controlled timing. The extended training cells implement the first two points. However, the earlier cells use a different procedure. They have been using a validation subset linked to the augmented training data. Checkpoint selection has been based on maximum validation accuracy and there is no clear indication that frozen batch-normalisation modules are explicitly kept in evaluation mode during training. The results have been labelled based on their stage.

The clean baselines have not been provided for the 500-image corruption subset. This means we cannot claim exact degradation for blur, noise, brightness and resolution. The tables contain the recorded results for the corrupted data. Comparisons between corruption settings use measurements from the same subset.

The notebook has been restarted, has out of order counters and has console logs pasted in some of the code cells. This is an exploratory notebook and the outputs are not guaranteed to be a clean run from start to finish. The dataset, trained weights and standalone predictions have not been provided within the experimental notebook. This document only provides evidence of what has been saved and does not claim to reproduce training from the missing files.

B. GENERALIZATION

Training and validation images have been split at image level using a fixed seed. No evidence has been provided to prove disjoint identities or that duplicates and near duplicates have been screened. This means we cannot make any stronger claims about disjoint identities from the index split. These checks would be useful before interpreting the results as robust generalisation.

The held-out directory has been inspected during initial, fine-tuned and extended evaluations, as well as corruption tests. The epochs have been chosen based on the validation. However, the broader development process has not been isolated from repeated observation of test results. The scores are best viewed as exploratory and will need external validation.

The experiments have only been run on one dataset, with one split and a limited corruption grid. There is no quantification of variability between different training seeds, image subsets and noise. The 500-image subset is smaller than the full test set. The 20-image explanation dataset is much smaller. Any small differences between the models have not been claimed to be statistically significant.

The RVF10K dataset has been used for real-versus-synthetic facial image classification. No evidence has been provided to prove generalisation to new generators, face swapped videos, temporal cues, audiovisual consistency, adversarial attacks, demographics or unrelated images. The results should not be claimed to generalise to all deepfake detection tasks.

C. TIMING AND EXPLANATIONS

The timing has been measured as a single run through the evaluation loop on the CPU for each model stage. This includes loading and preprocessing. There is no indication of a warm-up period or variability between runs. The processor model and memory or energy measurements are also unavailable. The JPEG and subset corruption evaluations use different batch sizes for inference. The results provide useful information for this run but cannot be used as a benchmark for hardware.

Grad-CAM is a coarse explanation method and depends on the selected layer. All four qualitative examples are fake images. One row has different target classes between the models. The consistency data does not contain any compressed predictions. These practical limits complement the concerns about saliency validation raised by Adebayo et al. [29]. The similarity statistics cannot prove the explanations are faithful, localisation is accurate or predictions are invariant. No extended checkpoint explainability has been evaluated.

D. FUTURE WORK

The extended ResNet18 will need to be evaluated using the same 3,000-image deterministic test. All five metrics and its confusion matrix will need to be provided. Both extended models will need to have matching clean subset and corruption evaluations. The previous robustness values cannot be used.

To make this reproducible, the split indices, checkpoint hashes, configurations, labels and prediction probabilities should be saved. By cleaning the notebook, there would be no reliance on hidden states and the tables and curves could be recreated. The seeds should be repeated to quantify uncertainty. The timing should be split into preprocessing and inference. Timing should include a warm-up and repeated measurements to quantify run-to-run variability. The explanations should include clean and corrupted predictions so similarity can be examined alongside correctness and class changes.

Generalisation to other datasets and new generators will need to be performed as the generalisation settings are different [14], [15], [16], [17], [18]. Corruption aware augmentation and threshold calibration could be looked into but would require new measurements to prove any benefits.

VII. CONCLUSION

This study evaluates lightweight real-versus-synthetic facial-image detection through clean classification, common image corruptions, computational measurements, and visual explanations. For the earlier fine-tuned checkpoints, MobileNetV3-Small achieves 83.13% clean accuracy and

ResNet18 83.87%. Their aggregate performance is close, but their error profiles differ: MobileNet detects a larger proportion of fake images, while ResNet18 produces fewer false alarms on real images.

MobileNet uses 1,518,881 parameters compared with ResNet18's 11,177,025 and has approximately 4.46 times the measured throughput in the recorded CPU evaluation loop. The efficiency advantage is accompanied by stronger default-threshold performance under severe JPEG compression and resolution loss. ResNet18 has higher accuracy under the tested Gaussian-noise levels. Both architectures fail to detect most fake images under severe blur, demonstrating why clean accuracy alone is insufficient.

Continued MobileNet fine-tuning with deterministic validation yields 85.63% held-out accuracy, 0.8596 fake-class F1, and 0.9355 ROC-AUC. Its improvement over the earlier checkpoint is mainly due to fewer real-image false positives. Extended ResNet18 reaches 91.24% internal validation accuracy, but the notebook contains no corresponding test evaluation, so an updated held-out architectural ranking remains unresolved.

Grad-CAM images and the complete JPEG-paired explanation record add diagnostic detail. MobileNet has mean heatmap cosine similarity 0.9302 and ResNet18 0.9017 across 20 pairs per model, but these values do not establish correctness or faithful causal explanations. The supported conclusion is that a compact detector can provide a useful accuracy-efficiency trade-off on the examined dataset, while still exhibiting substantial, condition-specific reliability limitations. A complete extended-checkpoint evaluation, matched subset baselines, controlled timing, and independent external tests are needed before stronger deployment claims.

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